

# A Closed Form for What the Model Emits: Template Anchoring in Unconditioned Zero-Shot Circle Packing

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*(Revision 2, 2026-08-01. Implements `p4_review_stats.md`, `p4_review_reviewer2.md` and `p4_review_gecco.md`, plus the arbitration in `p6_cruzes.md`. Every claim that changed relative to revision 1 is itemized in Table 5 (Appendix C); the deviations table is Table 4 (Appendix B), and the ledger correction, the review-process disclosure and the stopping rule are in §9.)*

## Abstract

Language models are increasingly placed in the proposal role of discovery loops such as FunSearch and AlphaEvolve, on the assumption that their outputs explore a diverse solution space. We characterize one component of that assumption: the *unconditioned* proposal — a single zero-shot, code-free call with no parent program, no fitness feedback and no evaluator in context. On a classic constructive-geometry benchmark, maximizing the sum of radii of  $N$  circles in a unit square, a weak-tier proposer does not search. It emits a grid-with-corner-fillers template and truncates it, even when a provably better construction is one parameter away. The behavior admits a closed form: a nearest-square order  $k^* = \text{round}(\sqrt{N})$  with a value function  $V(k, m)$  identifies the *empirical modal output* — which template, and hence its computed value — at all seven tested  $N$ , stated before sampling. Per-sample agreement equals that modal frequency — 56–86% by cell — while a round-number baseline hits 2 of 69 valid samples: the formula captures everything short of sampling entropy. We preregistered these point predictions with prompt hashes before sampling and tested them out of sample on two containers, a square and a rectangle, for which the rule was restated but never refitted; on the rectangle the informative result is negative — 0 of 11 valid samples reached the provably higher-scoring rival — while on-prediction support is partial (5 of 11, not separable from a uniform-template null). Across three tiers we find three attractor families and an inversion: constructive ambition rises monotonically with nominal tier while execution validity does not — it rises then collapses (78%  $\rightarrow$  100%  $\rightarrow$  13% at the primary  $10^{-6}$  tolerance; the third arm is an unattributable serving alias, reported with that caveat throughout). Trace elicitation is an intervention rather than an observation: requesting a method line concentrates outputs onto the anchor (87% vs 70% on-prediction,  $p = 0.03$  uncorrected — failing Holm over the registered family, carried by one of three cells and confounded with a collection-wave split of comparable size), with no detectable validity change ( $p = 0.30$ ,  $n = 60/\text{arm}$ ; not powered to exclude an effect of the observed size). Method lines are checkable against emitted coordinates: 54 of 56 scoreable claims (96.4%) describe the object actually built. Two scope conditions are load-bearing: the closed form is a weak-tier regularity — and a preregistered second-vendor arm (gemma-4-26b, §3.6) now confirms it at the pooled level, 57.5% of valid packings on the predicted sum against a registered 30% bar — while the cell-level prediction fell short at 2 of 5 scoreable cells matched (registered bar: 5; registered falsifier at  $\geq 4$  failures: not triggered), the two matches locking at 19/20 and 15/15 and every miss landing *above* the prediction — that does not describe the two higher tiers sampled; and it describes what an *unconditioned* call emits, not what a loop converges to. A preregistered extension run after external review (arm M, §3.4) sharpens the scope in both directions: the branch rule’s fifth  $k$  confirmed out of sample at  $N = 57$  (modal output  $T(8, 57) = 3.5625000$ , 0 of 10 valid samples on the rival), while the registered falsifier for the filler branch **triggered at 3 of 3 cells** — at  $N = 20, 30$  and 41 the model does not extend-and-fill but moves to the  $(k^*+1)$

grid, so  $V(k, m)$  describes behavior only on the  $m \leq 1$  support previously tested, and the extend branch of the rule is disconfirmed at  $m \geq 4$ . A second preregistered extension (arms MU and CH, §3.5) tested whether the result transfers to parent-conditioned calls, and its registered falsifier **also triggered**: given a provably better in-family parent, 0 of 26 valid mutation samples return to the anchor (keep-or-improve 26/26, grid-order inheritance 26/26), so the anchoring result is a property of unconditioned calls only in the in-family-parent regime — the one-parent regime that discovery loops actually run in is not described by it, and the title of this paper is scoped accordingly. The template reasserts itself only against a far-worse off-family parent (58.8% snap-back). A companion probe that enumerates the full recipe family in the prompt (arm CH) decomposes the mechanism: choice follows the stated score table, but execution fails off-template in 30 of 31 invalid attempts — the template is less what the model prefers than what it can reliably build.

## 1 Introduction

Ask a language model to place 21 circles in a unit square so as to maximize the sum of their radii, and it will not search. It reaches for the nearest square grid — five by five, radius  $1/(2k)$  — then truncates it to 21 circles, leaving four cells empty and their area unclaimed. Those cells are where the value went. A better construction is one parameter away: drop to a four-by-four grid and add corner fillers of radius  $(\sqrt{2} - 1)/(2k)$ , and the sum rises. The model does neither, and behaves the same way on the next sample.

The behavior can be written down. Writing  $k^*(N) = \text{round}(\sqrt{N})$  for the nearest-square order, a value function  $V(k, m)$  over grid order  $k$  and filler count  $m$  identifies the modal output of the proposal distribution — which template the model emits most often at each  $N$ , and hence its value, from the problem parameters alone, before sampling. The empirical mode equals the predicted value at all seven  $N$  tested, and per-sample agreement is bounded by the modal frequency itself, 56–86% by cell (§3.2). The rule also predicts where the behavior costs value: when  $k^{*2} \leq N$  the model extends the grid with fillers (a branch arm §3.4 later restricts to  $m \leq 1$ ); when  $k^{*2} > N$  it truncates instead of dropping to  $k^* - 1$  and filling. These *trap zones* —  $N \in [13, 15], [21, 24], [31, 35], [43, 48], [57, 63]$  (clipped to  $[57, 60]$  by our sweep bound) — are a property of the branch behavior, not of the packing problem.

Circle packing is the showcase benchmark of the LLM-driven discovery literature, the lineage FunSearch (Romera-Paredes et al., 2024) opened. It is the task on which AlphaEvolve (Novikov et al., 2025) reported 2.63586276 for  $n = 26$  and on which later systems report a narrow band — ShinkaEvolve (Lange et al., 2025) 2.635983283, HELIX 2.63598308 (Su et al., 2026), GigaEvo 2.636 (Khrulkov et al., 2025), AdaEvolve 2.636 (Cemri et al., 2026) — with ThetaEvolve (Wang et al., 2025) claiming new best-known bounds rather than matching it.

**What we measure, and what we do not.** Those systems query  $p(\text{program} \mid \text{parent programs} + \text{fitness scores} + \text{evaluator feedback} + \text{archive context, at generation } t > 0)$ . We query  $p(\text{coordinates} \mid \text{task description, no parent, no code channel, no feedback, generation } 0)$ . These differ on four axes at once — parent conditioning, output modality, feedback, generation index — and we do not claim to have measured the in-loop distribution. What we characterize is the *unconditioned* call. A source audit finds no loop stage in the cited systems that makes precisely this call: FunSearch reseeds emptied islands by cloning a best surviving program, ShinkaEvolve’s stagnation restart re-seeds from its own archive (“initial”, “best” or “archive\_random”), and OpenEvolve islands copy the user-provided seed — every in-loop call is program-conditioned. The nearest deployed relatives are initializations conditioned on a seed that “can be trivial” (FunSearch) or “rudimentary”, single-line functions returning constants (AlphaEvolve), plus AlphaEvolve’s “No evolution” ablation, which re-prompts from the same rudimentary program throughout. The unconditioned call is the limiting case these approach as seed information goes to zero, and finding that this limit is a template lookup, its value computable in advance, is actionable for the people who build these loops — seed diversification, forced- $k$  initialization, trap- $N$  avoidance in benchmark selection. Anything about what the loop samples at  $t > 0$  is a hypothesis here. The one published result bearing directly on the substitution — “Dictionaries, Not Darwin” (Li, 2026), reporting parent-conditioned evolution indistinguishable from fresh independent sampling at equal

budget — is warrant from equation discovery, not geometry. §8 names the experiment that would close the gap.

This paper makes three framing choices worth stating plainly. Behavioral anchoring is established territory (Huang et al., 2026; Lou & Sun, 2024; Malberg et al., 2025) and we inherit its vocabulary; what differs is modality — the anchor is a *construction template*, not a number. The benchmark choice is strategic; we did not pick it for novelty. Preregistration, too, is an adopted standard here, not a contribution in itself (Thomas et al., 2026; Williams et al., 2026; Vaccaro, 2026; Zhang, 2026); our twist is what is locked — exact-output point predictions from a closed form, on a held-out container.

**Contributions.** (1) *A closed form that names the modal model output in advance.*  $k^*(N)$  with  $V(k, m)$  and  $T(k, N)$  identify the value a weak-tier proposer emits most often at a given  $N$ , verified against a linear-programming oracle on 83 configurations to within  $10^{-9}$  and tested out of sample on two containers, the rectangle rule restated ( $q^* = \text{round}(\sqrt{N/a})$ ,  $p^* = \text{round}(\sqrt{N \cdot a})$ ) but never refitted. Prior work publishes functional forms predicting *aggregate* metrics ahead of a run (Kaplan et al., 2020; Hoffmann et al., 2022; Snell et al., 2024); to our knowledge none predicts a specific multi-decimal *individual emitted output value* ahead of sampling. The claim is scoped to the weak tier (§5). (2) *A tier boundary condition, as three attractor families.* Ambition rises monotonically with nominal tier; validity does not — 78%  $\rightarrow$  100%  $\rightarrow$  13% at the primary tolerance, 64%  $\rightarrow$  90%  $\rightarrow$  13% at  $10^{-9}$ . The most ambitious tier attempts recursive gaskets and mostly fails to produce a valid packing; it was addressed through a bare serving alias and appears as `opus_alias` with that caveat everywhere. (3) *Checkable faithfulness, and trace requests as interventions.* Method lines are checkable against emitted coordinates, and 54 of 56 scoreable claims (96.4%) describe the object actually built — a clean, well-powered finding (§6.5). Requesting the line at all is an intervention rather than an observation: the concentration shift it produced (87% vs 70%,  $p = 0.0325$  uncorrected) fails multiplicity correction, is carried by one of three cells and is wave-confounded (§6.4), so it is reported as met-as-registered exploratory material, not a demonstrated effect. Studies collecting process descriptors *by request* are nonetheless measuring a perturbed distribution.

**Non-claim guard.** Everything here is a behavioral regularity over emitted outputs. We make no claim about mechanism inside the weights, and the paper uses that vocabulary throughout: the model *emits*, the distribution *concentrates*, the formula *identifies the modal output* (storage-and-retrieval vocabulary corrected; Table 5, item 1). §8 states the leading mechanism-free alternative, arithmetic tractability, and records that we did not test it.

## 1.1 Trap zones

Substituting  $k^*(N)$  into the branch condition: TRAP  $N \in [k^2 - k + 1, k^2 - 1]$ , CONVERGE  $N \in [k^2, k^2 + k]$ . Over  $N = 10 \dots 60$ : [13, 15], [21, 24], [31, 35], [43, 48] and [57, 63] clipped to [57, 60] by the sweep bound.

Inside a zone the rule generally loses value against the best construction available *within the recipe family*. Worst-in-zone penalties fall as  $N$  grows: 8.51% at  $N = 13$ , 7.03% at 21, 6.01% at 31, 5.25% at 43, 4.66% at 57. The penalty reaches zero at  $N = 35$ , 48, 14 and 15, but **not** at  $N = 24$ , where truncation gives  $T(5, 24) = 2.4000000$  against  $V(4, 8) = 2.4142136$ , a residual 0.59% (Table 5, item 3). Branch label and penalty are therefore not coextensive: 4 of the 22 trap-branch  $N$  in range cost nothing, so “trap” names the branch, not a guaranteed loss. Where the penalty is zero, truncation is separable from convergence only by structure; §3.3 uses  $N = 35$  as that control.

Every closed-form value is recomputed over the constructed coordinates by an independent linear program that knows nothing about the recipe; the script aborts on disagreement. 83 configurations, both branches, every  $k$  in  $2 \dots 7$ , drift below  $10^{-9}$ .

Our bound table (`n_sweep_forecast.json`) covers  $N = 10 \dots 30$  only, and its  $N = 26$  entry, 2.63598, *is* ShinkaEvolve’s figure truncated, so the LLM-driven systems are the record on this problem (Table 5, item 4). What survives is that the recipe family is never competitive with that record: deficit 0.02–0.26 across  $N = 10 \dots 30$ , and 0.0946 at  $N = 26$  ( $2.63598 - 2.5414214 = 0.0945586$ ). Above  $N = 30$  there is no bound table, so both the deficit claim and the LP gate’s “never exceeds a published bound” abort are unchecked there, covering three of five trap zones and four of our seven square cells.

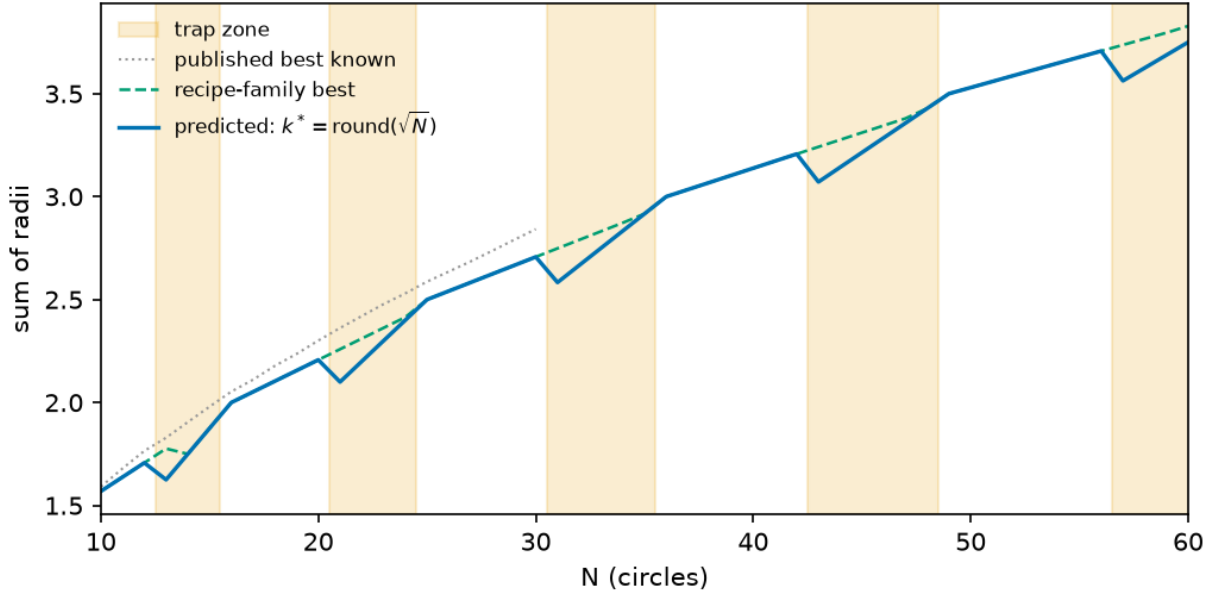


Figure 1: *Predicted versus optimal value*,  $N = 10 \dots 50$ . (i) the rule’s prediction; (ii) the best value in the recipe family; (iii, dotted) published best-known values, terminating at  $N = 30$  because the bound table stops there. Shaded bands are the trap zones, the last being  $[57, 63]$  clipped by `sweep(10, 60)`. Worst-in-zone penalties 8.51%, 7.03%, 6.01%, 5.25%; these are family-internal arithmetic and exact at every  $N$ , while the published-bound comparison (curve iii) is checkable only up to  $N = 30$ , where the bound table stops. The gap closes to zero at  $N = 35$  and  $N = 48$  but **not** at  $N = 24$ , where 0.59% remains — curve (ii) sits strictly above (i) across  $[21, 24]$ .

## 2 Task and recipe family

### 2.1 The benchmark

Maximize the sum of radii of  $N$  non-overlapping circles in the unit square. Feasibility is decidable exactly from emitted coordinates, so proposals score with no human in the loop.

All invocations are zero-shot and code-free. The prompt (Appendix A) fixes the count, states containment and non-overlap, forbids writing or executing code, and demands only a raw Python list of  $[\mathbf{x}, \mathbf{y}, \mathbf{r}]$  triples. Our reason for the code-free design is that a code channel routes the task to a numerical optimizer, so the distribution would reflect the optimizer. That reason is asserted, not measured — we ran no code-enabled arm (§8) — and it also means the measured object is not the object the code-channel systems query, which is why §1 scopes the claim to unconditioned calls.

### 2.2 The recipe family

Three terms, used consistently: the **recipe family** is the parametric set of grid-plus-filler constructions; a **template** is one member; the **anchor** is the template the distribution concentrates on at a given  $N$ .

A  $k \times k$  grid places one circle per cell center with  $r_{\text{grid}} = 1/(2k)$ ;  $m$  fillers sit on interior grid vertices, tangent to their four surrounding grid circles, with  $r_{\text{filler}} = (\sqrt{2} - 1)/(2k)$ ,  $0 \leq m \leq (k - 1)^2$ . Hence  $V(k, m) = k/2 + m(\sqrt{2} - 1)/(2k)$ . When  $N < k^2$  the observed behavior is not to drop to a smaller grid and fill it but to *truncate* — lay out the  $k \times k$  lattice, occupy  $N$  cells, leave the radius unchanged — giving  $T(k, N) = N/(2k)$ . These reproduce every previously reported anchor with no fitting:  $V(5, 1) = 2.5414214$ ,  $V(5, 2) = 2.5828427$ ,  $V(5, 0) = 2.500$ ,  $T(5, 23) = 2.300$ ,  $V(4, 7) = 2.3624369$ .

The finding that 94 of 95 valid proposals in prior arms were grid-plus-filler is retained only as motivation, since those arms are not reported here; the self-contained version is §5’s observation that across the 155 weak-tier rows collected before arm M only two exceeded the family best, both by  $\sim 10^{-7}$  (arm M’s 57 rows are scored separately in §3.4 and were not swept for this claim).

### 2.3 The order rule is a definition; the branch rule is the hypothesis

We write  $k^*(N) = \lfloor \sqrt{N} + \frac{1}{2} \rfloor = \text{round}(\sqrt{N})$  for the nearest-square order. This is a *definition* (Table 5, item 2). Given that the model emits some  $k \times k$  grid from this family, “the nearest square lattice” yields  $\text{round}(\sqrt{N})$  by construction; and  $\arg \min_k |k^2 - N|$  switches at  $N = k^2 + k + \frac{1}{2}$  where  $\text{round}(\sqrt{N})$  switches at  $k^2 + k + \frac{1}{4}$ , so on the integers the two coincide exactly — two formalizations of “nearest” agreeing is a symptom that no substantive hypothesis is being selected. Scoring four candidate orders against the three fitting anchors gave **nearest** 3/3, **floor** 2/3, **argmax** 2/3, **ceil** 1/3; under a null where each candidate matches each anchor with probability  $\frac{1}{2}$ , some candidate among four goes 3/3 about 40% of the time, so this is a weakly identified discrete selection, not a zero-parameter law.

The falsifiable content is the **branch rule**:  $k^{*2} \leq N \rightarrow \text{extend with } m = N - k^{*2} \text{ fillers (converge); } k^{*2} > N \rightarrow \text{truncate the } k^* \text{-grid (trap)}$ . The extend arm survived testing only at  $m \leq 1$  — a preregistered extension disconfirmed it at  $m \geq 4$ , where the model prefers the  $(k^*+1)$  rectangle grid (§3.4). Nothing about “nearest” forces truncation — a proposer with any local search would drop to  $k^*-1$  and fill, which is why the trap zones exist and why a higher-valued rival is reachable inside them. Truncate-versus-drop-and-fill is a genuine dichotomy, and it is what §3 and §4 test. The governing quantity is the signed distance  $N - k^{*2}$ . Primality is not the variable — 13, 23, 31, 43, 47 and 59 trap while 11, 17, 19, 29, 37, 41 and 53 converge — refuting the prior work’s conjecture about primes near 30 before any model is queried.

## 3 Preregistered forecast, out of sample

### 3.1 Registration protocol

For each cell the prompt is pinned and its SHA-256 digest written to disk first — the  $N = 13$  square prompt hashes to 32db485b.... Predictions P1–P5 are registered in the header of `arm_f_repro.py`, with standalone files for the parallel arms (`arm_s_preregistration.txt`, `arm_o_preregistration.txt`, `arm_t_preregistration.txt`). Raw outputs are stored verbatim, failures included; scoring is deterministic and local.

Four properties here are disclosed, not repaired. Temperature, top-p and top-k are not exposed by the runtime. The alias-to-weights binding is a promise, not a hash. The subagent inherits instruction files outside the task prompt, so the digests lock a *fragment* of the conditioning context — they are prompt-fragment hashes, and a replicator cannot reconstruct the inherited files from what we release. Finally, hash coverage is incomplete (§9).

Two scoring conventions were fixed in advance. Validity is reported at  $10^{-9}$  and  $10^{-6}$ , both logged,  $10^{-6}$  primary: proposers print six to eight decimals, so an eight-decimal tangency misses contact by  $\sim 5 \times 10^{-9}$ , while  $10^{-6}$  sits far below the  $\sim 10^{-2}$  gap between rival constructions. Both tolerances appear in every validity figure below, as registered. Value matching uses a  $2 \times 10^{-3}$  window; §5 records where that window is too loose for the distinction it was asked to carry.

**Reproducibility posture.** Prompt hashes were registered before sampling in the preregistration files for arms F, S, O and T (`arm_f_repro.py` header, `arm_s_preregistration.txt`, `arm_o_preregistration.txt`, `arm_t_preregistration.txt`); what is hashed is the task prompt only, not the harness, so the dispatch wrapper and the inherited instruction files sit outside the digest; what is not attestable at all is the `opus_alias` alias-to-weights binding, a vendor promise rather than a hash; hash-coverage gaps, the unhashed pilot prompt and the ledger provenance corrections are itemized in §9 and Appendix A.

### 3.2 The formula sits at the mode ceiling

An exact point prediction invites a fair objection: a per-sample hit rate well short of 100% is not “predicting the exact output”, which is what revision 1’s abstract claimed (Table 5, item 7). The number should be read against the sampling entropy of the distribution, computed in `p11_mode_baseline.json` (bare arm,  $2 \times 10^{-3}$  buckets). At **all seven tested  $N$  the predicted value is the empirical modal output**, and the on-prediction rate equals the modal frequency to within one sample:

| $N$ | predicted | empirical mode | modal freq | on-prediction / valid | $k^*$ -structure / valid <sup>†</sup> |
|-----|-----------|----------------|------------|-----------------------|---------------------------------------|
| 13  | 1.6250000 | 1.6250         | 10/18      | 10/18 (56%)           | 17/18                                 |
| 17  | 2.0517767 | 2.0518         | 3/4        | 3/4 (75%)             | 3/4                                   |
| 21  | 2.1000000 | 2.1000         | 12/15      | 12/15 (80%)           | 13/15                                 |
| 31  | 2.5833333 | 2.5833         | 12/17      | 13/17 (76%)           | 17/17                                 |
| 35  | 2.9166667 | 2.9167         | 3/4        | 3/4 (75%)             | 3/4                                   |
| 37  | 3.0345178 | 3.0345         | 3/4        | 3/4 (75%)             | 4/4                                   |
| 43  | 3.0714286 | 3.0714         | 6/7        | 6/7 (86%)             | 7/7                                   |

<sup>†</sup> Post-hoc structural check, not preregistered, prompted by an external review of this revision (`diagnostics_kmatch.py`): the grid order backed out of each sample’s dominant radius ( $k = \text{round}(1/2r)$ ) equals  $k^*(N)$ . All 50 of 50 on-prediction samples are  $k^*$ -structured — the value match is not an accounting coincidence — and 64 of 69 valid samples overall (93%) sit on the  $k^*$  grid, so most value misses are  $k^*$ -grid variants (perturbed radii or filler counts), not different constructions.

No predictor of a single value can exceed the modal frequency, and this one attains it at 7 of 7 cells (at  $N = 31$  the on-prediction count exceeds the 4-dp modal bucket by one sample — 13/17 against 12/17 — a bucketing convention, not a violation of the ceiling): the residual is dispersion, not misprediction. As a floor, a naive “the model emits a round number” baseline hits 2 of the same 69 valid samples (3%). Two scope conditions: this is the bare weak-tier arm — pooled over the four discriminating square cells (full ledger), the two higher tiers and the rectangle, the on-prediction rate is 46% ( $47/102 = 41/57 + 1/30 + 0/4 + 5/11$ ; the three non-discriminating square cells, 9/12, are excluded because prediction equals the family argmax there), because §5’s higher tiers are almost never on-prediction — and “modal” is a property of the observed sampling regime, not of a pinned decoding configuration (§8).

### 3.3 Square container

The square arm sampled Haiku-tier proposers at  $N = 13, 17, 21, 31, 35, 37$  and 43. The rule was fitted on  $N = 23/26/27$  only, so every cell is out of sample. Four cells are *discriminating*: rule and family optimum disagree, so a proposer that searched the family would return a higher number. Across those cells ( $N = 13, 21, 31, 43$ ), **18 of 23 valid invocations landed on the predicted construction** and the rival-argmax value was reached **2 times in 23** — both at  $N = 21$ , both the  $4 \times 4$  grid plus five fillers at 2.2588835.

Those figures cover the **original 45-invocation bare arm only** (ids  $\leq 5$  at  $N = 13/31$ ,  $\leq 10$  at  $N = 21$ , all rows at  $N = 17/35/37/43$ ), the corpus that existed when §3 was written. §6 later added 40 bare rows at  $N = 13/21/31$  under the same arm label; over the full shipped ledger the same four cells give **41/57 on-prediction and 2/57 rival**. Both are reported. §3 and §6 are computed on nested subsets of one arm, not on independent samples (Table 5, item 28), and Table 1 carries the split.

That the anchor first breaks at  $N = 21$  is suggestive — it is the bottom of a zone carrying a 7.03% penalty, so the anchor looks weak where obeying it costs — but that is a hypothesis, not a finding. **P4** registered  $N = 37$  as converging on 3.0345178, a prime predicted clean, contradicting the prior work’s guess; it confirmed. **P5** registered  $N = 35$  at 2.9166667, the top-of-zone control; three of four valid samples landed there, separable from convergence only because the structure classifier reads radii rather than totals; the fourth used a  $7 \times 7$  lattice truncated to 35 (sum 2.5), outside the rule. At  $n = 4$  this is “consistent with” rather than confirmed, and deserves running to  $n = 20$ .

Bookkeeping. Five invocations were rejected by the runtime’s 20-subagent concurrency cap *before reaching a model*. We excluded them under a rule that was not preregistered, so the arm is reported both ways —  $35/45 = 77.8\%$ , or  $35/50 = 70.0\%$  had the rejections been scored as model failures (Table 5, item 5) — and the five records appear in the ledger in no form (§9). The arm also contains three parse failures (Table 5, item 6), and one  $N = 37$  proposer derived  $r = (\sqrt{2} - 1)/12$  correctly in prose, then transcribed 0.03571429.

### 3.4 Preregistered extension (arm M): the filler branch fails at $m > 1$ ; the fifth $k$ confirms

Two external reviews of this revision observed, correctly, that (i)  $V(k, m)$ ’s filler branch was empirically supported only at  $m \in \{0, 1\}$  — the converge cells  $N = 17$  and  $37$  both have  $m = 1$  — and (ii) the  $k = 8$  trap zone appeared in Figure 1 but was never sampled. Arm M closes both gaps and was registered before sampling (`arm_m_preregistration.txt`, commit e528c6b; prompt hashes in `arm_m_prompts.json`):  $n = 15$  bare weak-tier invocations at each of  $N = 20$  ( $k^* = 4$ ,  $m = 4$ , predicted  $V(4, 4) = 2.2071068$ ),  $N = 30$  ( $k^* = 5$ ,  $m = 5$ ,  $V(5, 5) = 2.7071068$ ),  $N = 41$  ( $k^* = 6$ ,  $m = 5$ ,  $V(6, 5) = 3.1725890$ ) and  $N = 57$  ( $k^* = 8$ , trap,  $T(8, 57) = 3.5625000$ , rival  $V(7, 8) = 3.7366935$ ), with an explicit tie-stated falsifier for each half.

**The falsifier for the filler branch triggered at 3 of 3 cells.** The registered predictions P-M1–P-M3 are disconfirmed: not one of the three converge cells has  $V(k^*, m)$  as its modal output. The model does not extend a  $k^*$ -grid with fillers; it moves *up* to the  $(k^*+1)$  grid and truncates or exactly fills it — modal outputs  $2.0 = T(5, 20)$  (12 of 15 valid;  $N = 20$  is the  $5 \times 4$  rectangle exactly),  $2.5 = T(6, 30)$  (11 of 14;  $6 \times 5$  exactly) and  $2.9285714 = T(7, 41)$  (6 of 7;  $7 \times 6$  minus one). Exactly one sample in 36 valid converge-cell rows emitted the registered construction — a textbook  $V(4, 4)$ , all four fillers at  $(\sqrt{2} - 1)/8$ . Per the registered falsifier wording,  $V(k, m)$  is restricted to the  $m \leq 1$  support previously observed, and the branch rule’s extend arm is disconfirmed at  $m \geq 4$ ; the intermediate “places some fillers” outcome named in the registration did not occur (filler count 0 in 35 of 36 valid converge rows). The emergent regularity is arithmetic, not geometric: where  $N$  factors as  $a \times b$  with  $a, b$  near  $\sqrt{N}$  the model emits that exact rectangle of uniform circles ( $20 = 5 \times 4$ ,  $30 = 6 \times 5$ ), which is further evidence for §8’s arithmetic-tractability alternative — rectangle factorization is mentally computable where filler tangency is not.

**P-M4 confirmed.** At  $N = 57$  the modal valid output is  $T(8, 57) = 3.5625000$  (6 of 10 valid at  $10^{-6}$ ), the  $k = 8$  grid truncated to 57; 0 of 10 valid samples reached the rival  $V(7, 8)$ , against a registered rival prediction of 0. The branch rule’s truncate arm now has five confirmed  $k$  (4, 5, 6, 7, 8) and Figure 1’s  $k = 8$  zone is sampled. Validity by cell: 15/15, 14/15 (one parse failure — fraction literals), 7/12 and 10/15 at  $10^{-6}$ . Three  $N = 41$  invocations died in the runtime before reaching a model and are excluded under the rejection rule registered for this arm in `arm_m_preregistration.txt` — registered precisely because the analogous arm F exclusion was not (§9) — and counted here: the cell is  $n = 12$  sampled of 15 launched. Raw rows verbatim in `arm_m_collect.jsonl`; scoring in `arm_m_analysis.py`, conventions identical to arm F.

The net effect on the paper’s claim is a sharpening. Arm M does not rescue  $V(k, m)$ ; the closed form now reads “ $k^* = \text{round}(\sqrt{N})$ ” names the grid family; at  $k^{*2} > N$  the model truncates that grid (five  $k$  confirmed); at  $k^{*2} \leq N$  it extends only to  $m \leq 1$  — beyond that it prefers the  $(k^*+1)$  rectangle, and  $V(k, m)$  does not describe it.”

### 3.5 Preregistered extension (arms MU and CH): the anchor does not transfer to one-parent calls, and handing the model the argmax does not dislodge the template

Two questions were still open after arm M, both pressed by the council review of this revision. Does the unconditioned anchoring result say anything about calls that carry a parent program — the regime evolutionary loops actually run in — and does the §8 arithmetic-tractability alternative survive a direct probe? Arms MU and CH were registered together before sampling (`arm_mu_preregistration.txt`, commit 2b7d202; prompt hashes in `arm_mu_prompts.json`), with numeric decision thresholds, a power analysis (0.83 at the registered effect size), tie-against conventions, and a falsifier (F-MU1) whose consequence — the title and abstract of this paper are scoped to *unconditioned* calls only — was written down before the first invocation.

**Arm MU** issues one-parent mutation calls: the bare prompt plus an existing packing and the instruction to modify it so the sum of radii increases. Three parents per cell ( $N = 13, 21, 31$ ;  $n = 15$  per condition-cell, 135 invocations):  $A\_anchor$ , the model’s own unconditioned anchor  $T(k^*, N)$ ;  $B\_rival$ , the provably better in-family rival  $V(k^* - 1, m)$ ; and  $C\_offfamily$ , a diagonal-chain packing scoring  $\approx 0.59\text{--}0.62$ , far below either.

**F-MU1 triggered; the anchoring result dissolves under conditioning.** Under  $B\_rival$  the pooled anchor-rate is 0 of 26 valid (0%, CI  $[0.00, 0.13]$ ) against a registered DISSOLVES threshold of  $\leq 20\%$ ; keep-or-improve is 26 of 26 against a threshold of  $\geq 50\%$ . Every valid  $B\_rival$  sample keeps or refines the rival construction, and every one inherits the rival’s grid order ( $k$ -inheritance 26/26, disconfirming P-MU3’s  $< 50\%$  prediction). The template does not reassert itself when a better in-family parent is present.  $A\_anchor$  makes the same point from the other side: even with the model’s *own* anchor as parent and an explicit increase-the-sum instruction, only 7 of 28 valid samples (25%, CI  $[0.13, 0.43]$ ) stay on the anchor — P-MU1’s  $\geq 50\%$  prediction is not met. Per the registered falsifier wording: the unconditioned-call anchoring result does not transfer to parent-conditioned calls, the loop-relevance framing is disconfirmed for the one-parent regime, and this paper’s claims are scoped to unconditioned calls throughout.

The one condition where the template does reassert itself is the one the registration predicted it would: given the *off-family* diagonal parent, 20 of 34 valid samples (58.8%, CI  $[0.42, 0.74]$ ) abandon the parent entirely and emit a  $k^*$ -grid construction — P-MU4 holds per-cell at  $N = 21$  (7/13) and  $N = 31$  (12/14, every valid sample on the  $k = 6$  grid), though not at  $N = 13$  (1/7). The asymmetry is the finding: a good in-family parent is followed, a bad off-family parent is discarded for the template. The attractor is real but weak — it captures the trajectory only when the parent is worse than what the template supplies.

**Arm CH** hands the model the entire recipe family: the bare prompt plus every admissible  $T(k, N)$  and  $V(k, m)$  value enumerated, including the family argmax, with the instruction to use whichever scores highest or anything better ( $n = 15$  per cell, 45 invocations). The §8 tractability alternative says the model truncates at  $k^*$  because that is the only construction it can compute; enumerating the family removes the computation burden, so under that reading the modal output should become the stated argmax (P-CH2). Under the template reading it stays  $T(k^*, N)$  despite the argmax being printed in the prompt (P-CH1).

**Registered verdict: P-CH1 holds (2 of 3 cells) — but the registered metric carries a survivorship artifact we disclose rather than exploit.** Under the registered scorer, which takes the modal output *among valid samples*, the template reading wins: at  $N = 13$  the modal valid output is  $T(4, 13) = 1.6250000$  against a stated argmax of 1.7761424, at  $N = 31$  it is  $T(6, 31) \approx 2.5833$  (4 of 7 valid) against 2.7485281 (3 of 7) — a one-sample margin — and only  $N = 21$  goes to the argmax (2.2588835, 3 of 5 valid). P-CH2 is disconfirmed at 1 of 3 cells. Read no further and the tractability alternative fails the probe. But validity is low in this arm — 2, 5 and 7 valid of 15 by cell — and the failure modes reverse the picture at the *attempt* level: **all 31 invalid rows attempt the stated argmax** (every one carries the family’s filler radius). Of those, 30 misplace the fillers — typically at the container corners, where radius  $(\sqrt{2} - 1)/(2k)$  overlaps the adjacent grid circle; the correct construction places them at the grid interstices — and 1 emits radical literals the registered scorer cannot evaluate. Counting attempts rather than valid outputs, the model *chooses* the argmax in a large majority of CH invocations and simply fails to execute it; the valid-conditioned modal is dominated by the template because the template is the only construction the model executes reliably. The registered metric was fixed before sampling, so we report its verdict as registered — and flag in the same breath that it conditions on exactly the variable (execution success) that separates the two readings. The honest summary is a decomposition rather than a clean winner: handed the argmax, the model’s *choice* follows the score table, but its *execution* fails off-template in 30 of the 31 invalid attempts — the §8 tractability alternative is wrong about selection and right about instantiation. This is the same asymmetry arm MU found from the other direction, and it narrows the paper’s central claim to its defensible core: the template is not what the model prefers; it is what the model can build.

Raw rows verbatim in `arm_mu_collect.jsonl` (180 of 180 launched invocations collected, no run-time deaths); scoring in `arm_mu_analysis.py`, committed before sampling completed; frozen output in `arm_mu_results.txt`. This arm brings the study corpus to 468 invocations.



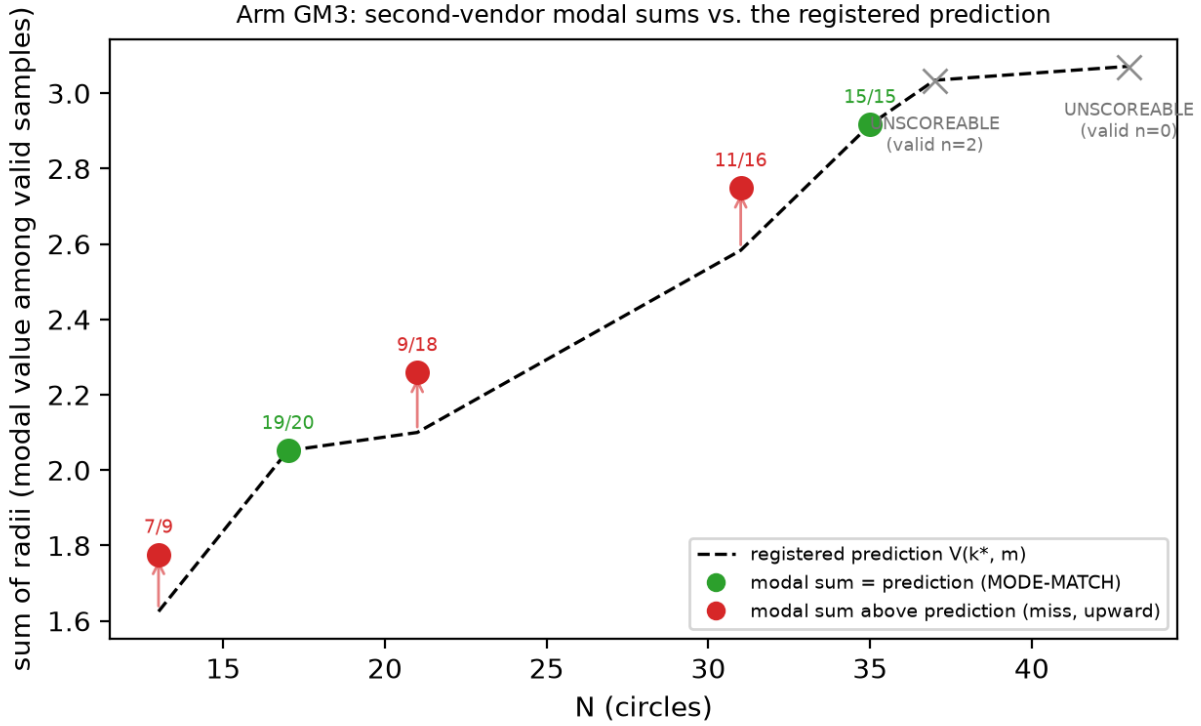


Figure 2: *Arm GM3 modal sums against the registered prediction.* Green cells match the predicted mode (frequencies annotated); red cells miss — every miss lands above the prediction; the two largest cells are UNSCOREABLE under the registered <3-valid rule. Regenerates via `fig_gm3_anchoring.py` from `arm_gm3_report.json` (itself replayed from the raw ledger by `arm_gm3_analysis.py`, no arguments).

### 3.6 Preregistered extension (arm GM3): the attractor crosses vendors at the pooled level, and departures from it point up

Every arm above addresses one vendor’s models. Arm GM3 asks whether the anchoring law is a fact about that lineage or about weak-tier models as a class, by re-running the unconditioned-call protocol against a second vendor’s open-weight family (gemma-4-26b-a4b-it) through its first-party API: seven  $N$  cells, twenty samples each, registered predictions and falsifier fixed before sampling (prereg chain in §9; the instrument’s history — a first attempt whose 4,096-token budget was consumed entirely by the model’s deliberation channel, resolved by a registered re-run at 16,384 — is in §8, and is itself a finding about output-budget confounds in cross-vendor comparisons). Per cell, recomputable from `arm_gm_gm3_checkpoint.jsonl` by `arm_gm3_analysis.py` with no arguments:

| $N$ | predicted (4 dp) | Gemma modal sum | mode freq | valid $n$ | on-prediction | MODE-MATCH         |
|-----|------------------|-----------------|-----------|-----------|---------------|--------------------|
| 13  | 1.6250           | 1.7761          | 7/9       | 9         | 0             | no (misses upward) |
| 17  | 2.0518           | <b>2.0518</b>   | 19/20     | 20        | 20            | <b>yes</b>         |
| 21  | 2.1000           | 2.2589          | 9/18      | 18        | 8             | no (misses upward) |
| 31  | 2.5833           | 2.7485          | 11/16     | 16        | 3             | no (misses upward) |
| 35  | 2.9167           | <b>2.9167</b>   | 15/15     | 15        | 15            | <b>yes</b>         |
| 37  | 3.0345           | —               | —         | 2         | 0             | UNSCOREABLE        |
| 43  | 3.0714           | —               | —         | 0         | 0             | UNSCOREABLE        |

Three registered outcomes. **The pooled prediction held with margin:** 46 of 80 valid packings (57.5%) land within  $2 \times 10^{-3}$  of the value predicted by §2’s rule, against a registered 30% bar — for a rule fitted entirely

on another vendor’s models, before this family was ever sampled. **The cell-level prediction did not:** the predicted value is modal in 2 of 5 scoreable cells, short of the registered 5, though the registered falsifier (failure in  $\geq 4$ ) also did not trigger — and where the family locks on, it locks on almost completely (19/20 at  $N = 17$ , 15/15 at  $N = 35$ ). **The structural signature came in under its bar:** 20 of 46 on-prediction samples (43%, bar: half) carry the exact two-radius grid-plus- filler decomposition, so on-prediction Gemma samples reach the predicted *value* through the predicted *construction* slightly less than half the time. The direction of the three cell-level misses is the informative part: every one misses *upward* — the family’s modal packing beats the registered construction (2.2589 against 2.1000 at  $N = 21$ ). The attractor this vendor’s family escapes toward is better packings, not noise, which is what the tractability reading of §8 predicts for a family with different arithmetic reach: the rule’s value marks where construction-by-mental-arithmetic lands, and a family that can execute slightly harder constructions leaves it in the only direction available. Validity collapsing at the two largest cells (2 then 0 valid of 20) bounds the family’s competence window on this task; both cells are UNSCOREABLE under the registered rule and claim nothing.

### 3.7 Preregistered extension (arm V): under the zero-shot protocol, anchoring transfers by vendor — and fails by family

Arm GM3 varied the vendor under the unconditioned-call protocol; arm V varies it under the zero-shot statement protocol of arm F, byte-identical prompts (SHA-256 per  $N$  restated in the registration and recomputed at call time, aborting on mismatch), against free-tier aliases across five vendors (`arm_v_preregistration.md`, committed before sampling; three amendments, each committed before the sampling it governs, expanded the original three-alias design to thirteen aliases across two serving paths — the opencode CLI and Open-Router chat-completions — and raised `max_tokens` to 24,576 after hidden-reasoning truncation emptied three models’ replies). Serving disclosures per the companion paper’s protocol: aliases on managed paths, `sampling_params`: null on every row, per-row upstream-provider strings logged where the path reports them, every invocation appended live to `arm_v_candidates_raw.jsonl` — 877 invocations, failures and refusals included. Scoring is `arm_f_repro.py`’s parse/validate/classify unchanged, tolerances registered in arm F before its sampling.

Registered floor:  $\geq 10$  of a model’s 25 cells valid at  $10^{-6}$ ; below it, no anchoring claim in either direction. Ten of thirteen aliases finished BELOW-FLOOR — including all three originally-registered opencode aliases, whose logged failure taxonomies are transport-dominated for one (CLI build banners, timeouts, “Unknown model” errors as the free tier rotated its catalogue) and roughly half transport, half parse-failure for the other two — plus one alias returning all 108 of its attempts without content. Three aliases cleared the floor, and their registered verdicts split:

| alias (vendor)                    | valid | V1 on-prediction | V1 verdict (bar: $\geq 50\%$ ) | V2 rival-argmax (bar: $\leq 10\%$ ) |
|-----------------------------------|-------|------------------|--------------------------------|-------------------------------------|
| north-mini-code (Cohere)          | 12    | 8/12 = 67%       | <b>TRANSFERS</b>               | 0/4 → <b>HOLDS</b>                  |
| gpt-oss-20b (OpenAI, open-weight) | 18    | 11/18 = 61%      | <b>TRANSFERS</b>               | 0/9 → <b>HOLDS</b>                  |
| gemma-4-31b-it (Google)           | 15    | 0/15 = 0%        | <b>DOES-NOT-TRANSFER</b>       | 0/3 → <b>HOLDS</b>                  |

This is the registration’s most informative branch pair. Anchoring under the zero-shot protocol is not an artifact of the original family — two further vendors’ models land the registered construction values at 61–67% of valid outputs, and the strong form holds everywhere evaluable: zero rival-argmax emissions in 16 discriminating-cell validities (the registration recorded the original family’s prior for the same statistic as 0/21). And it is not universal: gemma-4-31b, clearing the same floor, lands the prediction never. Its valid packings are unanchored and low (bests 0.88–1.85 against predictions 1.63–3.03) — not the rival, not the template, just weaker de-novo construction. Under the registration’s falsification map, the family-specific branch fires — any model clearing the floor with DOES-NOT-TRANSFER makes the closed form a family property, stated in §5 with data rather than a caveat — while the map’s all-transfer branch fails on gemma alone, so the reading that anchoring is an artifact of the original family is defeated by the same table that bounds its generality.

The gemma pair needs stating precisely, because this family now appears on both sides. Under GM3’s unconditioned protocol, gemma-4-26b pooled-HELD at 57.5% (the mixed-serving accounting; first-party-only, 59.0% — §8’s registered deviation applies to this figure); under arm V’s zero-shot statement protocol,

gemma-4-**31b** through a rate-starved alias (Google AI Studio upstream; 27–38 of each cell’s attempts returned as shared-pool 429s before content arrived) transfers nothing. Protocol, parameter count, and serving path all differ, so the pair bounds the family’s behaviour rather than contradicting itself — but the conjunction is exactly the shape §8’s tractability reading predicts: a family whose arithmetic reach differs from the original’s tracks the attractor under one elicitation and escapes it under another, and no single-protocol measurement would have seen the boundary.

## 4 Rectangle transfer

### 4.1 Transfer of the rule

For a  $1 \times a$  rectangle the template has two parameters:  $q^* = \text{round}(\sqrt{N/a})$  columns across the width,  $p^* = \text{round}(\sqrt{N \cdot a})$  rows across the height. At  $a = 1$  both collapse to  $\text{round}(\sqrt{N})$  — the same rule with the aspect ratio put back, not a new rule fitted to new data, and no rectangle model output existed when it was written. It was verified against an independent LP at 213 configurations, drift below  $10^{-9}$ , including a cross-domain check: at  $a = 1$  with  $p = q = k$  the square file’s closed form must equal this file’s LP. Shape mismatch grows as  $a$  leaves 1 — over  $N = 10 \dots 45$  the count of  $N$  whose predicted shape differs from the optimal one rises from 12 at  $a = 1.0$  to 23 at  $a = 1.5$ , 20 at  $a = 2.0$  and 23 at  $a = 3.0$ .

We probed the two sharpest cells with sixteen proposers in a container none had been given before:  $N = 19$  at  $a = 3$  (predicted 3.16666667, rival 3.5749194) and  $N = 25$  at  $a = 2$  (predicted 3.1250000, rival 3.4832492). With all sixteen scored, **5 of 11 valid proposals landed on the predicted value and 0 of 11 reached the rival**.

We characterize all eleven here, since a referee objected that the majority were left undescribed. On-prediction: two at  $N = 19/a = 3$  (3.166666667 and 3.16666673, the latter agreeing to six decimals not seven) and three at  $N = 25/a = 2$  (3.125 exactly). Off-prediction at  $N = 25/a = 2$ : two samples emitted a uniform  $5 \times 5$  grid at  $r = 0.1$  summing to 2.5000000 — the *square* rule with no aspect correction,  $k = \text{round}(\sqrt{25}) = 5$  — plus one at 3.0 and one at 3.151875. At  $N = 19/a = 3$ , **two** samples exceeded the prediction from outside the family, 3.45 and 3.5, both below the rival (Table 5, item 8).

We therefore report the rectangle as **partial out-of-sample support, not confirmation**:  $5/11 = 45\%$ , Wilson 95% CI [21%, 72%], and a proposer choosing uniformly among the three or so plausible template shapes would land on-prediction about a third of the time, so  $5/11$  does not separate cleanly from that null. Two qualifications: validity degrades in the tall container (4/8 at  $a = 3$  against 7/8 at  $a = 2$ , three of four  $a = 3$  failures being overlaps), a separate finding and a confound on that cell at  $n = 8$ ; and the rectangle ledger carries no prompt hashes, no run dates and no proposer fields, so the arm with the strongest transfer claim has the least provenance (§9).

### 4.2 Negative result: the closed form does not survive the move to rectangles

The rectangle generalizes the *rule* but not the *formula*, and we keep the failure because it is what a verification gate is for. In the square every interior vertex has four identical neighbors, so one expression covers every filler. In a  $1 \times a$  rectangle, with half-spacings  $h_x = 1/(2q)$  and  $h_y = a/(2p)$ , a filler is also capped by horizontal and vertical spacing to *adjacent fillers*. Those constraints are inactive when  $h_x = h_y$  — why the square case could not have revealed them — and bind only when neighboring vertices are occupied, so the cap depends on  $m$  and on which vertices a construction uses, and no expression in  $(p, q, m, a)$  reproduces it. The LP gate caught this on the first run: at  $a = 1$ ,  $p = 2$ ,  $q = 4$ ,  $m = 1$  the natural generalization  $\mathbf{rf} = \min(\mathbf{diag}, h_x, h_y)$  returns **1.125** against a true **1.1545085**, capping against a neighboring filler that does not exist. We retain closed forms only where provably exact (full grid and truncated grid) and use the LP as the value oracle for the extend branch, so what §4.1 tests is the LP-backed prediction.

Plain entries are the original 45-invocation bare arm (§3.3’s corpus); **bold** the same cells over the full 85-row bare ledger after §6 added 40 rows. † marks non-discriminating cells (prediction equals family argmax), excluded from rival denominators. Square discriminating cells: 18/23 and 2/23 on the original

Table 1: Forecast versus outcome, both containers.

| Cell        | Branch   | Predicted | Rival argmax | Valid / sampled             | On-pred / valid            | Rival |
|-------------|----------|-----------|--------------|-----------------------------|----------------------------|-------|
| $N=13$ (sq) | truncate | 1.6250000 | 1.7761424    | $4/5 \cdot \mathbf{18/20}$  | $3/4 \cdot \mathbf{10/18}$ | 0     |
| $N=17$ (sq) | extend   | 2.0517767 | †            | $4/5$                       | $3/4$                      | †     |
| $N=21$ (sq) | truncate | 2.1000000 | 2.2588835    | $7/10 \cdot \mathbf{15/20}$ | $4/7 \cdot \mathbf{12/15}$ | 2     |
| $N=31$ (sq) | truncate | 2.5833333 | 2.7485281    | $5/5 \cdot \mathbf{17/20}$  | $5/5 \cdot \mathbf{13/17}$ | 0     |
| $N=35$ (sq) | truncate | 2.9166667 | †            | $4/5$                       | $3/4$                      | †     |
| $N=37$ (sq) | extend   | 3.0345178 | †            | $4/5$                       | $3/4$                      | †     |
| $N=43$ (sq) | truncate | 3.0714286 | 3.2416246    | $7/10$                      | $6/7$                      | 0     |
| $N=19, a=3$ | truncate | 3.1666667 | 3.5749194    | $4/8$                       | $2/4$                      | 0     |
| $N=25, a=2$ | truncate | 3.1250000 | 3.4832492    | $7/8$                       | $3/7$                      | 0     |

corpus; 41/57 and 2/57 on the full ledger. Rectangle: 5/11 and 0/11. Sources: `arm_f_repro.py` over `arm_f_candidates_v2.jsonl`, `arm_g_rect.py` over `arm_g_candidates.jsonl`.

## 5 The tier ladder: three attractor families

Holding prompt, container and scoring fixed, we varied only the nominal proposer tier across the three cells that discriminate hardest between prediction and family optimum —  $N = 13, 21$  and  $31$ , each inside a trap zone. The tiers do not merely differ in success rate: they attempt qualitatively different constructions.

**The weak tier truncates templates.** The Haiku-tier proposer produces uniform grids of radius  $1/(2k)$  truncated to  $N$  circles, with corner fillers only when the grid underfills. Across the 45 original bare invocations spanning  $N \in \{13, 17, 21, 31, 35, 37, 43\}$ , **35 were geometrically valid at the primary  $10^{-6}$  tolerance (77.8%), 29 at  $10^{-9}$  (64.4%)** — recomputed from the ledger, correcting the headline validity figure and the inversion triple in the abstract and Contribution 2 (Table 5, item 9). At the three discriminating cells 12 of 16 valid samples landed on the predicted value and the rival was reached twice, both at  $N = 21$ .

**The middle tier perturbs and mixes.** The Sonnet-tier arm was preregistered in `arm_s_preregistration.txt`, disclosing that 5 of 20 samples at  $N = 13/21$  had been seen before registration and that  $N = 31$  was fully blind. It was valid 30/30 at  $10^{-6}$  and 27/30 (90%) at  $10^{-9}$  — so the “100%” leg is tolerance-dependent and both numbers are stated. It was almost never on-prediction: 0/10, 0/10, 1/10 — 1/30 pooled against the weak tier’s 12/16 at the same cells. Instead of truncating a uniform grid it perturbs one, with enlarged edge rows, hexagonal interior rows and two or three distinct radii in the same packing (29/30 multi-radius).

One Sonnet sample at  $N = 31$  emitted 27 circles at  $r = 1/12$  with 4 corner circles at  $r = 1/8$ , summing to 2.7499999991 — above 2.7485281, the best the family reaches at that  $N$ . Recomputed from its stored coordinates, the construction is tangency-tight: minimum pairwise and wall slack both exactly zero at tolerance zero, the exactly-tangent contacts involving the binary-exact  $r = 1/8$  corner circles while the  $r = 1/12$  circles carry small positive slack. It is the only sample in the study to leave the recipe family upward.

That sample also exposes a scoring defect. At  $N = 31$  the family rival (2.7485281) and this out-of-family value (2.75) differ by  $1.47 \times 10^{-3}$ , *inside* the registered  $2 \times 10^{-3}$  window, so the value classifier counted the escape as a hit on the construction it exceeded. The window was chosen for the anchor comparison, where the nearest competing value is  $\sim 0.15$  away, and reused where it does not fit. Classified by structure instead, Sonnet’s rival count at  $N = 31$  is 2/10 and pooled 5/30 (Table 5, item 10). Every categorical claim in this paper about “reached the rival” or “left the family” is made at structure level or at  $10^{-6}$ , never at  $2 \times 10^{-3}$ .

**The top tier attempts recursive constructions and fails to build them.** The third arm was invoked through a bare tier alias. We name it `opus_alias` throughout and make no claim about which weights served it: the runtime accepts only the bare alias and exposes no dated identifier, so the binding is a vendor promise rather than an attestable fact. No version number appears anywhere in this paper.

Table 2: Three-attractor ladder.

| Tier                 | Cells                 | $n$ | Attempted family                                                        | Valid $10^{-6}$ | Valid $10^{-9}$ | On-pred                 | Rival         | Failure mode                      |
|----------------------|-----------------------|-----|-------------------------------------------------------------------------|-----------------|-----------------|-------------------------|---------------|-----------------------------------|
| haiku (bare)         | 13,17,21, 31,35,37,43 | 45  | uniform grid, truncated; fillers when underfull                         | 35 (78%)        | 29 (64%)        | 12/16 (3 matched cells) | 2             | overlap 6, outside 1, parse 3     |
| haiku, matched cells | 13,21,31              | 60  | as above                                                                | 50 (83%)        | —               | 35/50                   | 2             | overlap 5, outside 2, parse 3     |
| sonnet               | 13,21,31              | 30  | perturbed hexagonal 2–3 radii                                           | 30 (100%)       | 27 (90%)        | 1/30                    | 5 (structure) | none                              |
| opus_alias ‡         | 13,21,31              | 30  | quarter-circle corners, Apollonius fillers, coarse grid + border strips | 4 (13%)         | 4 (13%)         | 0/4                     | 0             | overlap 24, non-positive radius 2 |

That caveat matters because of two anomalies. Completion times ran 2.8–9 s across all 30 invocations against 75–250 s for Haiku and 150–1170 s for Sonnet, and the reported token count was uniform at 49,906 across the first 20 completions,  $\approx 49.9\text{k}$  thereafter — a signal of metadata that is not per-completion, or of a cache, router-fallback or truncation path. **Neither anomaly appears in any released artifact** — we therefore treat both as anecdotal context from an unreleased session transcript, not as data-supported properties of the tier: the ledger has no latency and no token field, and both figures come from the runtime session transcript, so a referee cannot check them.

Under those caveats the arm — preregistered fully blind in `arm_o_preregistration.txt` — was valid in 4 of 30 invocations (13%) at both tolerances. At every cell it attempted a more ambitious construction than either other tier: quarter-circle corners at  $r = 0.25$  with Apollonius-style fillers at  $N = 13$  (10/10 samples), mixed-radius  $4 \times 4$ -ish grids at  $N = 21$ , and a coarse  $3 \times 3$  grid at  $r \approx 1/6$  with border strips at  $N = 31$  (0/10 valid). Failures are geometric rather than numerical — edge strips at  $r = 0.03$  placed 0.138 from an  $r = 1/6$  grid circle needing 0.197 — and twice the arm padded to count with zero-radius circles. Valid samples score *below* the trap they were expected to fall into (1.26–1.41 at  $N = 13$  against 1.625). A post-hoc diagnostic (not preregistered; labeled as such in the repository) finds no evidence of a tolerance artifact behind the 13% in this sample: 24 of 26 invalid samples overlap grossly (median maximum overlap  $3.3 \times 10^{-2}$ ; radius-shrink repair costs a median 15% of sum-of-radii), while tolerance-scale near-misses ( $< 2.5 \times 10^{-5}$ ) occur instead in 5 of 7 geometry-scored weak-tier failures — the exact-tangency grids, not the ambitious constructions.

Two decisions here were made after seeing the data and are recorded as deviations (Table 4). P-O1, P-O2 and P-O4 are reported **not evaluable**, on the grounds that a validity collapse of this size makes an on-prediction tier comparison uninterpretable; P-O3 is trivially satisfied on 4/4 valid samples; the registered disconfirmation — regression toward the trap — did not occur. De-registering three of four predictions in the one arm that disconfirmed is exactly the move preregistration exists to constrain, which is why it is tabled rather than narrated. And the arm was initially excluded from the ladder, then included once its results were known; we follow the later decision and report the ladder both ways — with the arm  $78\% \rightarrow 100\% \rightarrow 13\%$ , without it  $78\% \rightarrow 100\%$  plus the qualitative shift from truncation to perturbed multi-radius hybrids, which is already the publishable contrast.

‡ Addressed through a bare serving alias; the alias-to-weights binding is not attestable, and this row is not a statement about any released model version. Tier denominators are not matched (the weak tier spans seven  $N$  against three), which is why the matched-cell row is given separately, and it is the matched comparison ( $83\% \rightarrow 100\% \rightarrow 13\%$ ) that should be quoted across tiers. Sources: `arm_f_candidates_v2.jsonl`, the three preregistration files.

“Monotone” applies to one axis only (Table 5, item 11): ambition rises monotonically — truncated template, perturbed hybrid, recursive gasket — while validity rises then collapses. “Ambition” is measurable, not merely a qualitative read: a post-hoc diagnostic (disclosed; `diagnostics_ambition.py`, all samples with

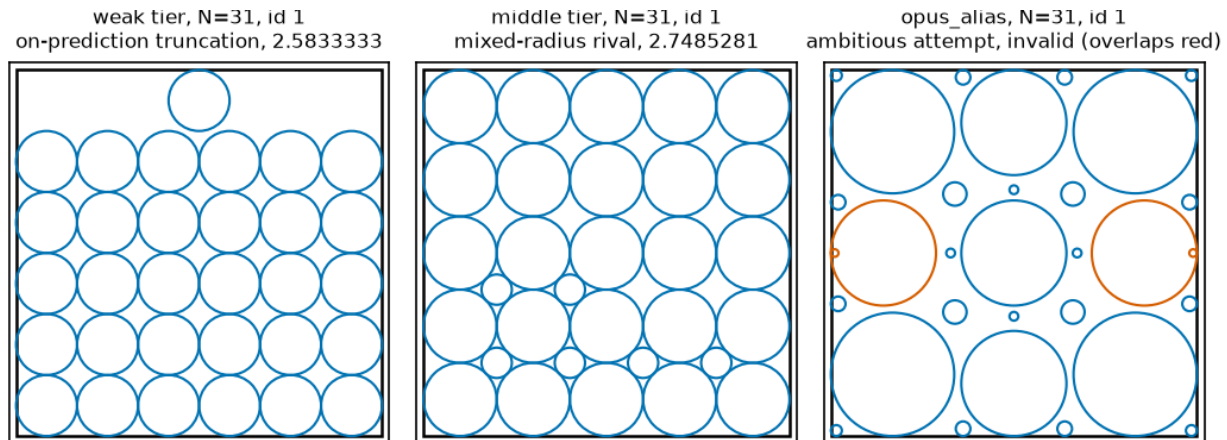


Figure 3: *Three attractor families at a single cell,  $N = 31$ .* One representative packing per tier, ledger sample id in each panel title (weak tier **bare** id 1, on-prediction truncation 2.5833333; middle tier **sonnet\_bare** id 1, mixed-radius rival 2.7485281; **opus\_alias** id 1, invalid, with only the offending circles highlighted). Source: `arm_f_candidates_v2.jsonl` via `fig_scripts.py`, deterministic (the exemplar rows’ geometry is byte-identical in v1; v2 differs only in provenance metadata).

emitted geometry, invalid included) gives median distinct radii per sample  $1 \rightarrow 2 \rightarrow 3$  (means  $1.33 \rightarrow 2.40 \rightarrow 3.37$ ) and median off-lattice center fraction  $0.08 \rightarrow 0.43 \rightarrow 0.95$  across the three tiers — monotone on both operationalizations. At the canonical and plausibly contaminated cell  $N = 26$  all tiers converge on the same 2.5414 attractor; they diverge only at withheld trap cells. The branch rule of §2 is therefore a weak-tier regularity — cross-vendor under both protocols (pooled under GM3’s, at 61–67% of valid outputs in two further vendors under arm V’s, §3.6–3.7), family-bounded (absent in gemma-4-31b at the same floor), cell-level modes confirmed in one lineage — and this is a **boundary condition on the main result**, not an independent second finding. It also cuts against §1’s framing: the discovery systems cited there run frontier or mid-tier proposers, so the tier at which our regularity holds is the tier they do not use, while the tier closest to what they use escapes the trap in our own data.

## 6 Elicitation and faithfulness

This section is explicitly secondary material and the result leans negative: the registered validity prediction fails, the concentration effect survives only as a fragile directional finding, and the faithfulness audit checks description against artifact, not against process.

### 6.1 Design and bundling disclosure

A ten-versus-ten pilot at  $N = 21$  compared the bare prompt against a variant asking the proposer to name its construction on a leading **METHOD**: line: validity rose from 7/10 to 10/10 and the higher-scoring rival disappeared, from 2 of 7 valid bare samples to 0 of 10 trace samples. Before any scaled sample was drawn we registered four predictions and an explicit falsifier in `arm_t_preregistration.txt` (sha256 `ab7900a8...`), disclosing that the pilot’s trace prompt had drifted from the bare template beyond the method line — it omitted the `[0,1]x[0,1]` tokens and reworded the output-format line — so the pilot’s intervention was method-line-plus-rewording, bundled. Pilot samples are never pooled with `trace_v2`.

The scaled prompt, `trace_v2`, is itself a **bundled prompt-format-and-trace-request** intervention, though a near-minimal diff: one inserted **METHOD**: line, plus "After the METHOD line, " prepended to the output line and "no other text" changed to "no other text after the list". That second change is not cosmetic — §6.2 shows it doing measurable work — so the measured effect is *method-line-plus-output-line-*

Table 3: Paired trace grid (scaled arm only).

| $N$           | Arm      | $n$ | Valid (parse / geom. fail) | On-pred / valid | Rival |
|---------------|----------|-----|----------------------------|-----------------|-------|
| 13            | bare     | 20  | 18 (0 / 2)                 | 10 (56%)        | 0     |
| 13            | trace_v2 | 20  | 18 (0 / 2)                 | 16 (89%)        | 0     |
| 21            | bare     | 20  | 15 (1 / 4)                 | 12 (80%)        | 2     |
| 21            | trace_v2 | 20  | 18 (0 / 2)                 | 16 (89%)        | 0     |
| 31            | bare     | 20  | 17 (2 / 1)                 | 13 (76%)        | 0     |
| 31            | trace_v2 | 20  | 17 (0 / 3)                 | 14 (82%)        | 1     |
| <b>pooled</b> | bare     | 60  | 50 (3 / 7)                 | 35 (70%)        | 2     |
| <b>pooled</b> | trace_v2 | 60  | 53 (0 / 7)                 | 46 (87%)        | 1     |

*rewording*, not the method line alone: a weaker version of the confound for which the pilot was discarded, held to the same standard.

The scaled arm ran 100 new invocations, bringing both arms to 20 per cell at  $N \in \{13, 21, 31\}$  and the corpus to 215 logged square invocations (85 bare = 45 original + 40 new; 70 trace = 10 pilot + 60 trace\_v2; 30 sonnet; 30 opus\_alias), plus 16 rectangle invocations, 231 total at the close of revision 2 (Table 5, item 27); the post-review arm M (§3.4) later added 57 sampled weak-tier invocations (60 launched), bringing the study corpus to 288, and the post-council arms MU and CH (§3.5) added 180 more (180 launched, none lost), for a final corpus of 468. One disclosure revision 1 did not carry: **20 of the 60 bare samples are pre-existing arm-F rows** ( $N = 13$  ids 1–5,  $N = 21$  ids 1–10,  $N = 31$  ids 1–5) whose results were known when P-T1–P-T3 were written; only trace\_v2 is fully fresh (§6.4). Scoring used `arm_f_repro.py` unchanged with the registered  $2 \times 10^{-3}$  window; the Fisher exact test comes from the hypergeometric tail in `arm_t_analysis.py`, checked against a reference.

## 6.2 Registered outcomes

Preregistered criteria are evaluated exactly as registered, and unregistered alpha thresholds are applied nowhere; revision 1 applied one asymmetrically (Table 5, item 12).

**P-T1, validity: NOT confirmed.** P-T1 is the one prediction that registered an alpha (“validity  $\geq$  bare at EACH  $N$ , and pooled one-sided Fisher  $p < 0.05$ ”). Direction held at all three cells, but the pooled test gives  $p = 0.30$ : 53/60 (88%) trace\_v2 against 50/60 (83%) bare, a difference whose detection needs several hundred samples per arm at 80% power — a failure to detect, not a demonstration of no effect (Table 5, item 13). The direction that did hold is not geometric: bare had **3 parse failures and 7 geometric failures**, trace\_v2 **0 parse and 7 geometric**, so among parsed samples geometric validity is 50/57 (87.7%) versus 53/60 (88.3%). The P-T1 direction is format compliance, exactly what the reworded output-format line manipulates; validity is reported split by cause from here on.

**P-T2, rival suppression: CONFIRMED as registered** (directional, no alpha): 1 of 53 valid trace\_v2 against 2 of 50 valid bare. Counts this near zero carry no inferential weight and we claim none (Fisher  $p = 0.48$ , carrying no decision). One trace\_v2 sample at  $N = 31$  hit the rival 2.7485281 to within  $3.6 \times 10^{-8}$  — the first weak-tier rival hit at  $N = 31$  in any arm, and formerly one of the scorer’s mismatches (§6.5).

**P-T3, anchor concentration: met as registered (directional), inferentially fragile.** Among valid samples, 46 of 53 trace\_v2 (87%) landed on the registered prediction against 35 of 50 (70%) bare. The  $p$ -value it never carried is 0.0325, one-sided uncorrected (§6.4).

**P-T4, faithfulness: confirmed under a corrected scorer.** The registered scorer gives 38 of 41 scoreable claims matching (93%) against a 90% threshold; blind hand-adjudication under a rubric frozen before rescoring gives 54 of 56 (96.4%). §6.5 reports both.

Per-cell one-sided Fisher on on-prediction:  $N = 13$   $p = 0.030$ ;  $N = 21$   $p = 0.41$ ;  $N = 31$   $p = 0.50$ . Pooled  $p = 0.0325$ ; Holm threshold over the registered family ( $m = 3$  tested with  $p$ -values) is 0.0167. The pilot (10

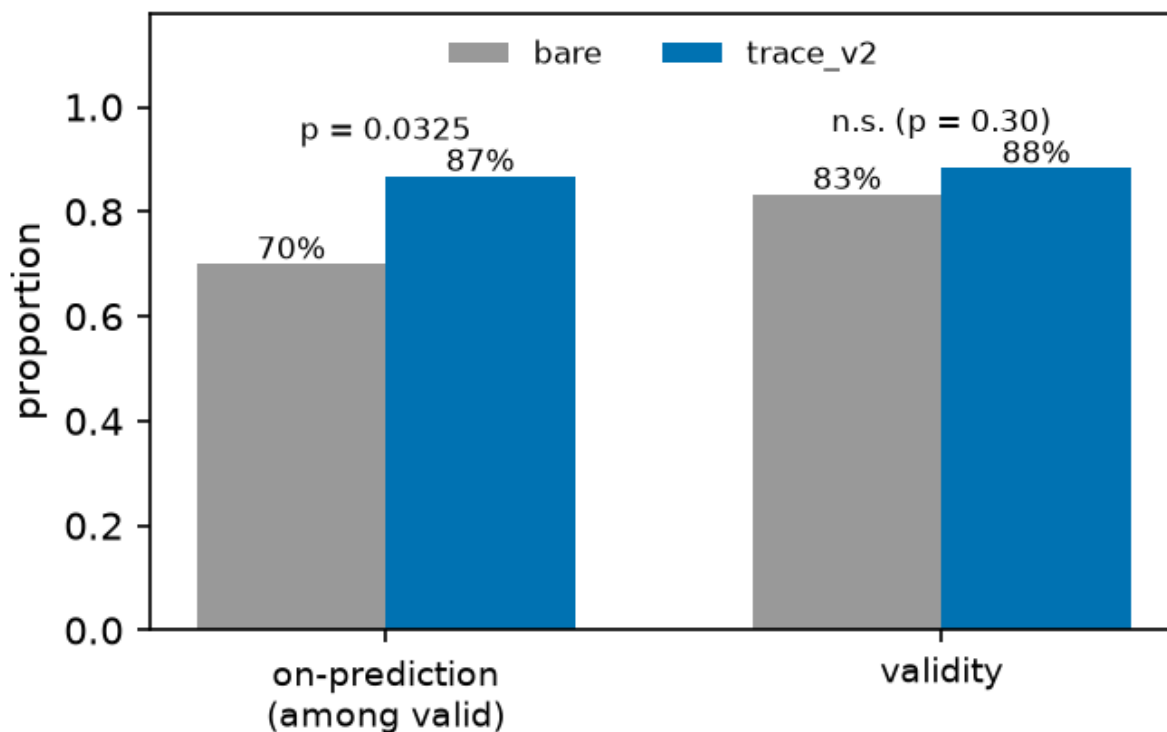


Figure 4: *Bare versus trace\_v2, by cell and pooled.* Caption caveat: the bars pool the bare arm’s two collection waves, which §6.4 shows drift materially, so the pooled bare bar is a mixture; the same-wave comparison in §6.4 is the wave-controlled picture.

invocations at  $N = 21$ , 10/10 valid, 9/10 on-prediction) is reported separately and never summed into any total. Regenerated by `arm_t_analysis.py` over `arm_f_candidates_v2.jsonl`.

### 6.3 The registered falsifier, both readings

The registered falsifier reads: “if `trace_v2 validity`  $\leq$  `bare` at 2+ of 3  $N$ , the pilot effect was the bundled rewording, not the method-line request — reported as such, not reframed.” Observed validity:  $N = 13$ , 18/20 both arms — tie;  $N = 21$ , `trace_v2` 18/20 against `bare` 15/20 — trace higher;  $N = 31$ , 17/20 both arms — tie. A tie satisfies  $\leq$ , so under the registered wording the falsifier is **triggered at 2 of 3  $N$** , and we report it as triggered; the code instead evaluated direction as `t_rate`  $\geq$  `b_rate`, strict  $<$ , under which it fires at 0 of 3 cells, which is what revision 1 reported (Table 5, item 14). That convention appears nowhere in the preregistration, was chosen at analysis time, and read the same tie favorably twice — as “direction held” for P-T1 and as “not a falsifier cell” here.

We name the failure mode **operationalization drift between preregistration text and analysis code**: registered prose leaves boundary behavior — ties, inclusive versus exclusive bounds, rounding — implicit, and the scorer must choose after the data exist; the preregistration literature (§7) discusses what is locked, not this tie-handling layer. Remedies, free before sampling and impossible afterward: register the comparison as executable code, or state the tie convention in the registration text. `arm_t_analysis.py` now prints both readings and names the registered tie-inclusive one authoritative. Under that reading **the pilot’s validity effect is attributed to the bundled rewording, not to the method-line request**, as the falsifier clause instructed; the §6.2 parse-versus-geometric split points the same way, and P-T1 fails under both readings regardless.



## 6.4 Limits of P-T3: what the concentration result will and will not carry

P-T3 met its registered directional criterion. It is not a flagship result.

*No registered alpha.* P-T3 registered direction only; the  $p$ -value is a post-hoc addition.

*Multiplicity.* Of four registered predictions, three are reported with  $p$ -values. Holm sets the first threshold at  $0.05/3 = 0.0167$ ;  $p = 0.0325$  fails it, as it fails Bonferroni at any  $m \geq 2$  and Benjamini–Hochberg FDR at  $q = 0.05$ . Two-sided Fisher is  $p = 0.0537$ . P-T3 is nominally significant uncorrected and is not under family-wise error control.

*Concentration in one cell.* Per-cell tests give  $p = 0.030, 0.41, 0.50$  (Table 3); the pooled 17-point gap is a 33-point gap at  $N = 13$  averaged with 9- and 6-point gaps elsewhere, driven by a *low bare rate at  $N = 13$*  ( $10/18 = 56\%$ ) rather than a high trace rate — the trace rate there (89%) equals that at  $N = 21$ . Dropping  $N = 13$ , the comparison is  $30/35$  vs  $25/32$ ,  $p = 0.31$ .

*Wave confound.* The bare arm mixes two collection waves; `trace_v2` is one. Splitting bare on the registered id boundary, with no intervention applied to either half, on-prediction is:  $N = 13$ ,  $3/4$  old wave vs  $7/14$  new;  $N = 21$ ,  $4/7$  vs  $8/8$ ;  $N = 31$ ,  $5/5$  vs  $8/12$  — the control arm’s own between-wave drift is comparable to the effect attributed to the manipulation. (An earlier draft printed  $2/4$  vs  $10/11$  at  $N = 21$  and a  $25/37$  pooled figure — arithmetically impossible against §6.1’s id boundary; corrected here from the ledger.) Restricting to one wave, new-wave bare is  $23/34$  (68%) against `trace_v2`’s  $46/53$  (87%), but **this stratified comparison is post hoc, not preregistered, no confirmatory weight** — a diagnosis, not a rescue: wave is a confound *in the registered analysis*, which cannot separate it from the manipulation. The design fix, interleaved collection in one session, was not run.

*What we therefore claim.* A minimal method-naming request does not make the proposer better at geometry and does not measurably change which rare constructions it reaches; in the registered direction it concentrates output onto the anchor, but that concentration fails correction, is carried by one cell, is confounded with collection wave, and is bundled with an output-format rewording that demonstrably affects parse compliance. Studies collecting process descriptors *by asking for them*, including descriptor-driven quality-diversity pipelines where the descriptor is a requested self-report, should treat this as a reason to check, not a coefficient to apply. The pilot also showed a *third* effect in the P-T3 direction at the same cell (`trace`  $9/10$  on-prediction against bare  $N = 21$  at  $4/7$ , one-sided  $p = 0.16$ ), so P-T3 replicated a pilot signal rather than testing a blind prediction — a different status, and the prereg’s pilot disclosure covered the prompt drift only (Table 5, item 15).

## 6.5 Faithfulness with ground truth

Claim and artifact sit in the same completion and the artifact is fully determined, so a method line is checkable against the emitted coordinates. `arm_f_repro.trace_faithfulness()` matches numeric dimensions from the method line against the emitted layout signature, returning 38 of 41 scoreable claims matching (93%) against the registered 90% threshold. Its 12 exclusions are regex coverage gaps, not claims without numeric content (Table 5, item 16): `DIMS_RE` matched only the literal  $N \times N$  form and `ROWSCOLS_RE` required rows before columns, so these fell through despite carrying explicit checkable dimensions — “Regular 3 by 4 grid with one additional circle in row 4”; “3 complete rows of 4 circles and 1 centered circle in the top row”; “rectangular grid 4-4-4-1 with uniform radius  $1/8$ ”; “Four-row grid with 4 circles in the first row and 3 in each of the remaining”; “Rectangular grid arrangement with 5 columns and 4 rows, plus one additional circle on top edge”; “Square grid 5 columns 4 rows with 1 additional circle at top”; “Rectangular grid six columns by six rows radius one-twelfth with five circles removed”; “Regular 5-column by 6-row grid plus one additional circle in the right margin”; “Rectangular grid packing with five columns and six rows plus one additional circle”. Three further exclusions — “Uniform horizontal strips with tight row spacing”, “Regular rectangular grid packing with corner circle”, “Equal-radius rectangular grid with 4 rows stacked vertically” — are closer to genuinely unscorable. The excluded set is enriched for non-canonical phrasings, where a mismatch is *a priori more likely*, so the exclusion was not conservative in a known direction; worst case, had all 12 been unfaithful, the rate would have been  $38/53 = 72\%$ .

**Blind rescore under a frozen rubric.** The rubric (`p7_faithfulness_rubric.md`) — checkable assertions in any surface form, plus match rules including a base-grid convention for “grid plus additions” claims and a truncation convention for “with k removed” claims — was committed to git as `e181d2a` at 03:56:56 on 2026-08-01, **before** any rescoring run, so the freeze is checkable, and it pre-committed the reporting rule: below 90%, P-T4 reported not confirmed, full stop, no consolation framing. Scoring was blind — claim text and circles only, no predicted or rival values, no running tally, no threshold and no sight of the existing 38/41 result — with tallying afterward; all 60 labels are released verbatim in `p7_blind_labels.json`.

**Result.** Over all 60 `trace_v2` rows: **54 MATCH, 2 MISMATCH, 4 UNSCOREABLE** — a corrected rate of  $54/56 = 96.4\%$  of scoreable claims, Wilson 95% CI [88%, 99%], passed, and on 56 scoreable claims rather than 41. The CI’s lower bound sits just below the registered threshold, so “clears 90%” is a statement about the point estimate at this  $n$ . The two mismatches share a form: row 11 claims “alternating rows of 4 and 3 circles” over observed row sizes 4,3,3,3, row 33 claims “5 rows alternating between 5 and 4 circles” over observed 5,4,4,4,4 — both *mis-descriptions of alternation*, a repeating pattern named and built for the first period only. The original scorer’s three mismatches were three different claim forms, not one (Table 5, item 17); the filler-adds-rows mechanism revision 1 described was real, and the rubric’s base-grid convention handles it, so all three score MATCH under hand adjudication.

**Scope, three ways.** The blind pass covers *invalid* rows too: the original 93% conditioned on completions that produced a correct packing — the wrong conditioning, since a method line attached to an overlapping layout is where description and artifact are likeliest to diverge — while the blind pass scored all 60 rows including the 7 invalid ones, 4 of the 60 unscoreable. Second, the check verifies the description against the *artifact*, not the process: the non-claim guard applies in full, and for a model already emitting grids, “ $5 \times 5$  grid” matching a  $5 \times 5$  grid is closer to a self-consistency check than to the causal question the chain-of-thought literature asks (§7). Third, the audit is confounded with the elicitation arm, computed on the arm §6.2 shows concentrates 87% of valid outputs onto one template, where correct description is close to free; faithfulness is informative precisely on off-template outputs, such as the pilot’s triangular-hexagonal  $6+5+4+3+2+1$  sample, described exactly while summing to 1.75. The split by on- versus off-prediction is not run here (§9, stopping rule). What survives is narrow: in a domain where claims are checkable, the method lines this proposer emits are, at 54 of 56 scoreable claims, true of the object it produced — and both failures are a model describing a pattern it started and did not finish.

## 7 Related work

Citation errors in revision 1 are corrected here and itemized in Table 5 (items 18–26).

**LLM-driven discovery and the circle-packing scoreboard.** Language Model Crossover (Meyerson et al., 2024), ELM (Lehman et al., 2022), EvoPrompt and LLaMEA (van Stein & Bäck, 2025) established the LLM call as an EC operator over prompts and programs; FunSearch turned the pattern into a discovery claim, and every system in §1’s scoreboard inherits it. Two orderings in that scoreboard matter: HELIX’s figure is marginally *below* ShinkaEvolve’s while being described as state of the art, and ThetaEvolve claims new best-known bounds, so the benchmark is saturated only across the AlphaEvolve–ShinkaEvolve–HELIX cluster and the number is still moving. OpenEvolve, the open-source AlphaEvolve reproduction most practitioners run, reports  $\approx 2.635977$ . §1.1 records the correction to our record comparison and its  $N > 30$  coverage gap. Two critiques bear on our framing and are frequently conflated, so we attribute them explicitly: Gideoni et al. (2026) show simple baselines recover much of the reported advantage, and Berthold et al. (2026) show classical solvers do too.

**Template convergence and diversity collapse.** “Mutation Without Variation” (Gurkan et al., 2026) finds iterated LLM program mutation collapsing onto previously seen structural templates — in 87% of chains, over 93% of mutations revisit a previously seen form — which is the same bias family in a different regime: mutation loops, program space, no closed form, no preregistration. On anchoring, Huang et al. (2026) treats anchoring as a general bias over initial information while Lou & Sun (2024) and Malberg et al. (2025) document numeric primes dragging point estimates; our modality is the difference, since a construction template has a computable value where a scalar anchor has only a measurable pull. Concurrently, “Measuring the Gap Between Human and LLM Research Ideas” (Chen et al., 2026) reports the same collapse one level up,

in idea space: across 11,683 paired proposals, LLM research ideas concentrate on bridge-and-synthesis framings (47–64% versus 12% for humans; normalized entropy down to 0.55 versus  $>0.92$ ), and chain-of-thought pushes the distribution *further* from the human one. That result is distributional; ours is the limiting case where the collapsed distribution’s mode is a single closed-form-predictable object — narrowness measurable there, computable here. “Artificial Hivemind” (Jiang et al., 2025) documents the same homogeneity across open-ended domains at survey scale, which makes cross-domain generality of the collapse the expected default; against that backdrop our contribution is not that collapse occurs but that, in this regime, its mode is a single closed-form-computable object, stated before sampling. Converging skepticism about what the proposer contributes comes from “Dictionaries, Not Darwin” (Li, 2026), Wang et al. (2026), BehaveSim (Zhang & Lu, 2026), Strategy Diversity (Yang et al., 2026), the bin-packing critiques (Herrmann & Pallez, 2026; Sim et al., 2025) and MathConstruct (Balunović et al., 2025). Two results cut the other way and belong on the other side of the ledger: Zhang et al. (2024) provides empirical grounding for the *importance* of evolutionary search, which is evidence against our substitution of an unconditioned call for the loop and is engaged in §8; and Zhang et al. (2026b) finds strong LLM optimizers act as *local refiners*, which is what a parent-conditioned call would be, so it bears on §8’s mutation arm.

**Scaling inversions.** Zhou et al. (2024) show larger and more instructable models becoming less reliable; the o3/o4-mini system card shows the same on PersonQA; Shojaee et al. (2025) reports collapse past a complexity threshold; GeoBuildBench (Kim et al., 2026) finds geometric construction a regime where nominal capability and executed correctness diverge. Our instantiation is *attractor family* versus executability.

**Trace faithfulness, and what our result is not.** One cluster intervenes *on trace content* and finds answers do not move — Reasoning Theater (Boppana et al., 2026), Project Ariadne (Khanzadeh, 2026), “Beyond the Commitment Boundary” (Scalena et al., 2026) and “The Chain Holds, the Answer Folds” (Li et al., 2026) all report causal decoupling. A second cluster *estimates* faithfulness without observing the process, represented here by Arcuschin et al. (2026); §6.5 reports the case where ground truth for the described artifact exists. Our result sits at a third intervention point: we vary whether a trace is *requested* rather than editing its content, which is measurement reactivity, not causal faithfulness. The closest cousin is “The Price of Format” (Yun et al., 2025), showing format constraints collapse generation diversity — and as §6.1 concedes, our manipulation includes a small output-format change, so we commit a miniature version of the confound we cite. Format-restriction performance costs are documented at scale in Tam et al. (2024). Related: (Abdelnabi & Salem, 2025; Zhang et al., 2026a; Wu et al., 2025).

**Preregistration lineage.** Thomas et al. (2026), Williams et al. (2026), Vaccaro (2026) and Zhang (2026) preregister recipes, outcome-blinded predictions and agent protocols; HindsightBench (Jia, 2026) releases frozen preregistrations of directional aggregate hypotheses. We adopt the standard rather than claim it. What we add is the object locked — exact closed-form point predictions of specific output values — and, from §6.3, the observation that locked prose and executing scorer can disagree at the boundary. Structural precedents: Kaplan et al. (2020) and Hoffmann et al. (2022) publish a functional form then run the confirming instance; Snell et al. (2024) is the closest structural twin. QDAIF (Bradley et al., 2023) is the QD-descriptor setting most affected by our §6 result, and GigaEvo (Khrulkov et al., 2025) the concrete in-scope system running MAP-Elites with LLM-driven mutation, where a self-reported behavioral descriptor would be perturbed in practice.

## 8 Limitations and untested alternatives

**The measured distribution is not the in-loop distribution.** We sample generation-0, parent-free, code-free, feedback-free calls; the systems in §7 sample parent-conditioned program proposals under selection. The claim that survives is about unconditioned calls, and Zhang et al. (2024) is published evidence that evolutionary search contributes materially, cutting against any stronger reading. The decisive cheap experiment is a one-parent mutation arm: same cells, prompt = bare template plus one parent packing plus its score plus “propose a modification that increases the sum of radii”, under three parent conditions (the on-prediction truncated grid, the higher-value family rival, an off-family parent), measuring whether the branch rule still predicts and whether the model inherits the parent’s  $k$ . That experiment has now been run as the preregistered arm MU (§3.5), and its registered falsifier triggered: 0 of 26 valid rival-parent samples return

to the anchor, keep-or-improve and  $k$ -inheritance are both 26/26, and the anchoring result is scoped to unconditioned calls as a stated negative transfer result. The concern this paragraph raised in revision 2 was correct.

**A mechanism-free alternative, now directly probed.** Under an explicit “construct the packing by reasoning alone” constraint, a  $k \times k$  grid at  $r = 1/(2k)$  is close to the only construction whose coordinates can be emitted from mental arithmetic without error accumulation, and  $\text{round}(\sqrt{N})$  is the arithmetically nearest such grid. That hypothesis predicts every observation in §2–§3 without appeal to stored templates, and predicts §5’s inversion too: the more ambitious tier attempts constructions whose coordinates it cannot compute by hand and overlaps — precisely the observed failure mode (24 overlaps and 2 nonpositive radii out of 26 `opus_alias` failures). The separating experiment — hand the model the recipe family explicitly, state  $V(k, m)$  and the admissible  $k$ , and ask it to *choose*  $k$ ; if it picks the argmax, the tractability reading stands — has now been run as the preregistered arm CH (§3.5). The outcome splits the hypothesis in two: the model’s *choice* follows the stated score table (so tractability does not explain the selection), but its *execution* fails off-template in 30 of 31 invalid attempts (so tractability does explain the instantiation — the grid is what survives mental arithmetic). §1’s vocabulary remains behavioral and no memorization claim is made.

**Contamination is not probed.**  $N = 26$  is the canonical published cell and plausibly in training data. We ran no canary-string test, no comparison against  $N$  values absent from the literature, and no perturbed-container test preserving difficulty while destroying lexical overlap. The rectangle arm is the nearest thing we have, at  $n = 11$  valid.

**Vendor scope, and what the cross-vendor chain cost to learn.** All tiers in the main study come from one provider; §3.6 is the arm that moves the pooled-level claim across vendors. The instrument’s history matters as much as its verdict and is recorded here. Arm GM (gemini-2.5-flash-lite, prereg commit 37b3adb) stalled on first-party quota at 55 of 140 and was completed through a routed serving path under a registered amendment (`arm_gm_amendment1_openrouter.md`, provider logged per row — every resumed row reports Google as the upstream). Its registered verdict is UNDERPOWERED in both accountings: 20 valid packings of 140, three scoreable cells against the five the primary requires, pooled on-prediction 15% against the 30% bar, falsifier not triggered (`arm_gm_analysis_v2.py`, `arm_gm_v2_report.json`). Two descriptives, claimed as nothing more: the one mid-size scoreable cell matches the predicted mode exactly (2.1000 at  $N = 21$ , 3/6), and off-prediction valid sums land far *below* the prediction (0.48–1.55) — the opposite direction from GM3’s misses, consistent with §5’s point that the regularity is tier-bound rather than universal. Arm GM2 (gemma-4-26b-a4b-it, prereg 3019aab) completed 140/140 invocations with 0 of 140 parseable — and revision 2’s two candidate readings (format inability versus budget truncation) were separated by GM3 as registered: the vendor API returns deliberation as a separate thought channel, GM2’s 4,096-token budget died entirely inside that channel, and at 16,384 tokens the same model parses at 118/140. A cross-vendor null under a fixed output budget is not evidence about the model until the budget’s interaction with the vendor’s response structure is itself checked. One mechanical deviation on GM3 is on record (amendment registered before resumption): the final 41 of 140 cells were collected through a routed third-party serving path after the first-party quota stalled, provider logged per row, both accountings computed — the deviation moves the pooled rate from 59.0% (first-party only) to 57.5% (mixed) and changes no verdict. Scope after §3.6: the anchoring law is cross-vendor at the pooled level; its full cell-level mode structure remains confirmed only within the original family. Arm V (§3.7) extends the chain under the zero-shot protocol: registered before sampling with three pre-sampling amendments, 877 invocations across thirteen free-tier aliases, ten below the registered floor with their failure taxonomies logged, and the three floor-clearing vendors splitting TRANSFERS/TRANSFERS/DOES-NOT-TRANSFER (`arm_v_score.py`, no arguments, from the live ledger; frozen output `arm_v_score_final.txt`).

**Sampling parameters unpinned.** The runtime does not expose pinned decoding parameters, so every effect is a distributional claim over the observed sampling regime, and §3.2’s mode-ceiling result inherits that scope. This is the subject of the companion paper.

**opus\_alias provenance.** The label denotes the serving alias addressed, not an identified model version. Without pinned weights we cannot separate a model-tier effect from a serving-path effect, and the supporting anomalies live only in the runtime transcript.

**Confirmatory base for the branch rule, updated by arm M.** floor and round disagree exactly on  $N \in [k^2 - k + 1, k^2 - 1]$ , so every discriminating cell is a trap cell by construction. The truncate arm now has five confirmed  $k$  in the square (4, 5, 6, 7, 8 — the last added by arm M’s  $N = 57$  cell, §3.4) plus two rectangle cells; the extend arm is confirmed only at  $m \leq 1$  and **disconfirmed at  $m \geq 4$**  (§3.4), which is the sharper scope the abstract now carries. The empirical- $k$  back-out named as future work in earlier revisions has since been run twice as disclosed post-hoc diagnostics: `diagnostics_kmatch.py` (§3.2 — 50/50 on-prediction samples  $k^*$ -structured, 64/69 valid overall) and arm M’s per-cell  $k_{\text{emp}}$  distributions (§3.4 — the converge cells sit on  $k^*+1$ , which is how the filler-branch failure was diagnosed).

## 9 Reproducibility, deviations, and disclosures

Deviations from preregistration are tabled in Appendix B; every claim that changed relative to revision 1 is tabled in Appendix C. Provenance, in list form:

- **Registration.** Prompts were hashed with SHA-256 and the hashes recorded before sampling, with the predicted values they were to be tested against: `arm_f_repro.py` (header, P1–P5, square arm), `arm_s_preregistration.txt`, `arm_o_preregistration.txt`, `arm_t_preregistration.txt` (prereg hash `ab7900a8...`), `arm_m_preregistration.txt` with `arm_m_prompts.json` (committed `e528c6b` before sampling), and `arm_mu_preregistration.txt` with `arm_mu_prompts.json` (committed `2b7d202` before sampling; decision thresholds, power analysis, tie conventions and falsifier F-MU1 all registered), and the cross-vendor chain `arm_gm_preregistration.txt` (commit `37b3adb`), `arm_gm2_preregistration.txt` (`3019aab`), `arm_gm3_preregistration.txt` with its serving-path deviation registered in `arm_gm3_amendment1_openrouter.md` before the affected cells were sampled, and `arm_v_preregistration.md` (commit `5444f10`, ancestry-checked by its runners before any sampling) with its three amendments, each committed before the sampling it governs. The rectangle transfer was registered before collection and never refitted. On multiplicity: this study now spans many registered arms, and no correction across arms is applied — each arm’s predictions were registered with its own bars and falsifiers before its sampling, every arm is reported regardless of outcome (three GM-chain instruments failed or stalled before one succeeded, and §3.4’s falsifier fired), and no arm was selected for inclusion on its result. The protection against garden-of-forking-paths here is registration and exhaustive reporting, not a family-wise error rate; a reader who wants one can count arms from this section.
- **Four limits on that claim.** (i) The digests cover the task prompt only; the subagent inherits instruction files outside it, so what is locked is a *fragment* of the conditioning context, a replicator cannot reconstruct those files, and we cannot establish they were identical across the arm-F and arm-T windows — the boundary §6.4 identifies as a confound. (ii) `arm_f_prompts.json` covers  $N = 13, 17, 31, 35$  and  $37$ ; the  $N = 21$  bare hash is registered in `arm_t_preregistration.txt`; the  $N = 43$  **bare hash appears in no preregistration and no prompts file**, existing only in the ledger — it recomputes exactly from the bare template, but recomputation after the fact is not registration. (iii) **No hash was ever registered for the pilot prompt**, whose drift the preregistration describes only in prose; the ten pilot rows carry `prompt_sha256: null` in the corrected ledger with an explanatory note. (iv) The rectangle ledger (`arm_g_candidates.jsonl`) carries no prompt hashes, no run dates and no proposer fields at all.
- **Ledger metadata correction.** Three referee findings established that the shipped ledger’s provenance fields contradicted the paper: trace rows carried the *bare* prompt hash for their cell, `run_date` was uniformly `2026-07-30` on all 215 rows, and the Sonnet and `opus_alias` rows were stamped with the Haiku alias and dated id. All three are corrected in `arm_f_candidates_v2.jsonl`; the correction is metadata only — every scored quantity, geometry, raw output, arm label and sample id is byte-identical to v1, verified field by field, each corrected field carrying a sibling `*_v1` field. `arm_f_candidates.jsonl` is unmodified (sha256 `02ecabcd...`); v2 is `9f845f78...`. Correction rules, sources, wave-boundary arithmetic and caveats — including that `run_date` has day granularity only — are in `ledger_v2_corrections.md`. Analysis scripts read v2.

- **Excluded and unreleased records.** The five concurrency-cap rejections appear in the ledger in no form, no `excluded_reason` field exists, and the exclusion rule was not preregistered; both rates are reported in §3.3. The `opus_alias` latency and token-count anomalies (§5) come from the runtime session transcript; no latency or token field exists in any released artifact.
- **Analysis artifacts.** Raw outputs are stored verbatim (`arm_f_raw.json`, `arm_f_candidates.jsonl`, `arm_f_candidates_v2.jsonl`, `arm_g_candidates.jsonl`, `arm_m_collect.jsonl` scored by `arm_m_analysis.py`, `arm_mu_collect.jsonl` scored by `arm_mu_analysis.py` with frozen output `arm_mu_results.txt`, and the GM3 ledger `arm_gm_gm3_checkpoint.jsonl` — every row’s raw API response verbatim, serving path and provider tagged per row — scored by `arm_gm3_analysis.py` into `arm_gm3_report.json` with both serving-path accountings) and scoring is deterministic and local. Value claims are gated on an independent LP oracle rather than on the closed form being tested: `n_sweep_forecast.py` with `verify_against_lp` checks 83 configurations to within  $10^{-9}$ , and the same oracle produced §4.2’s negative result. The faithfulness rubric (`p7_faithfulness_rubric.md`) was frozen by git commit `e181d2a` (2026-08-01 03:56:56) before any rescoring, and all 60 blind labels with justifications are in `p7_blind_labels.json`. `arm_t_analysis.py`, `n_sweep_forecast.json`, `rect_forecast.json`, `p11_mode_baseline.json` and `STATE.md` are included; every table regenerates from raw outputs by a command documented in `HOW_TO_RUN.md`.
- **Use of AI systems.** This paper studies language-model behaviour and was written with language-model assistance; both facts are stated here so a reader can weigh them together. Claude models (principally the Opus, Fable and Sonnet tiers) wrote the collection harnesses, the scoring, analysis and verification scripts, and the structural-check tooling, and drafted the manuscript prose; the external referee reports came from `deepseek-v4-pro`, `deepseek-v4-flash` and Gemini under written protocols (next bullet). The human author directed the programme, set the research questions, approved each preregistration before sampling, made the final inclusion, stopping and submission decisions, and is solely responsible for the content; no language model is an author. The claim a reader can check is the ordering, not the authorship: each preregistration commit is a git ancestor of the sampling it governs. The drafting of those preregistration texts, like the rest of this repository, was model-assisted. Two consequences follow from that, and neither is cosmetic. First, the models that wrote the analysis are of the same family as arms under study (§5), so the blind-spot risk flagged below for the referee panel applies to the analysis code as well — the LP oracle, the frozen scorer output and the referee panel’s independent recomputations exist in part to bound it. Second, every number here was produced by a script a model wrote, which is why the scripts, the raw ledgers and the frozen outputs are all released: what a reader is asked to trust is the artifact, not the author.
- **Review process.** The three referee reports that drove this revision (`p4_review_stats.md`, `p4_review_reviewer2.md`, `p4_review_gecco.md`), the arbitration of their conflicting recommendations (`p6_cruxes.md`), and the blind faithfulness adjudication of §6.5 were produced by language models under written protocols, not by human referees. They independently recomputed the paper’s statistics from the released ledger and found eight recomputation mismatches, all itemized in Appendix C, and their arithmetic was checkable and was checked. But the panel is not peer review, its verdicts carry no external authority, and it shares a family resemblance with the system under study — a blind-spot risk disclosed here rather than left implicit.
- **Stopping rule.** This revision closed after implementing the three named referee reports and the arbitration verdict; no further data-dependent analysis was performed. Analyses named but not run — the one-parent mutation arm, the recipe-family choice probe,  $N = 57$ , the empirical- $k$  back-out, faithfulness split by on- versus off-prediction, a code-enabled probe, interleaved re-collection of the arm-T pair, and P5 at  $n = 20$  — are recorded in §6.5 and §8 as future work, so that the reported statistics remain the ones the preregistrations governed.

## References

Sahar Abdelnabi and Ahmed Salem. The hawthorne effect in reasoning models: Evaluating and steering test awareness, 2025. URL <https://arxiv.org/abs/2505.14617>.

- Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthooan Rajamanoharan, Neel Nanda, and Arthur Conmy. Chain-of-thought reasoning in the wild is not always faithful, 2026. URL <https://arxiv.org/abs/2503.08679>.
- Mislav Balunović, Jasper Dekoninck, Nikola Jovanović, Ivo Petrov, and Martin Vechev. Mathconstruct: Challenging llm reasoning with constructive proofs, 2025. URL <https://arxiv.org/abs/2502.10197>.
- Timo Berthold, Dominik Kamp, Gioni Mexi, Sebastian Pokutta, and Imre Polik. Out-of-the-box global optimization for packing problems: New models and improved solutions, 2026. URL <https://arxiv.org/abs/2605.04850>.
- Siddharth Boppana, Annabel Ma, Max Loeffler, Raphael Sarfati, Eric Bigelow, Atticus Geiger, Owen Lewis, and Jack Merullo. Reasoning theater: Disentangling model beliefs from chain-of-thought, 2026. URL <https://arxiv.org/abs/2603.05488>.
- Herbie Bradley, Andrew Dai, Hannah Teufel, Jenny Zhang, Koen Oostermeijer, Marco Bellagente, Jeff Clune, Kenneth Stanley, Grégory Schott, and Joel Lehman. Quality-diversity through ai feedback, 2023. URL <https://arxiv.org/abs/2310.13032>.
- Mert Cemri, Shubham Agrawal, Akshat Gupta, Shu Liu, Audrey Cheng, Qiuyang Mang, Ashwin Naren, Lutfi Eren Erdogan, Koushik Sen, Matei Zaharia, Alex Dimakis, and Ion Stoica. Adaevolve: Adaptive llm driven zeroth-order optimization, 2026. URL <https://arxiv.org/abs/2602.20133>.
- Ziyu Chen, Yilun Zhao, and Arman Cohan. Measuring the gap between human and llm research ideas, 2026. URL <https://arxiv.org/abs/2607.01233>.
- Yonatan Gideoni, Sebastian Risi, and Yarin Gal. Simple baselines are competitive with code evolution, 2026. URL <https://arxiv.org/abs/2602.16805>.
- Can Gurkan, Forrest Stonedahl, and Uri Wilensky. Mutation without variation: Convergence dynamics in llm-driven program evolution, 2026. URL <https://arxiv.org/abs/2606.05408>.
- Julien Herrmann and Guillaume Pallez. An in-depth study of llm contributions to the bin packing problem, 2026. URL <https://arxiv.org/abs/2510.27353>.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models, 2022. URL <https://arxiv.org/abs/2203.15556>.
- Yiming Huang, Biquan Bie, Zuqiu Na, Weilin Ruan, Songxin Lei, Yutao Yue, and Xinlei He. Understanding the anchoring effect of llm with synthetic data: Existence, mechanism, and potential mitigations, 2026. URL <https://arxiv.org/abs/2505.15392>.
- Haozhe Jia. Hindsightbench: A black-box behavioral audit protocol for parametric hindsight in time-indexed llm decision tasks, 2026. URL <https://arxiv.org/abs/2607.18867>.
- Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond), 2025. URL <https://arxiv.org/abs/2510.22954>.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. URL <https://arxiv.org/abs/2001.08361>.
- Sourena Khanzadeh. Project ariadne: A structural causal framework for auditing faithfulness in llm agents, 2026. URL <https://arxiv.org/abs/2601.02314>.

- Valentin Khrulkov, Andrey Galichin, Denis Bashkirov, Dmitry Vinichenko, Oleg Travkin, Roman Alferov, Andrey Kuznetsov, and Ivan Oseledets. Gigaevo: An open source optimization framework powered by llms and evolution algorithms, 2025. URL <https://arxiv.org/abs/2511.17592>.
- Jinwoong Kim, Rui Yang, and Huishuai Zhang. Geobuildbench: A benchmark for interactive and executable geometry construction from natural language, 2026. URL <https://arxiv.org/abs/2605.13167>.
- Robert Tjarko Lange, Yuki Imajuku, and Edoardo Cetin. Shinkaevolve: Towards open-ended and sample-efficient program evolution, 2025. URL <https://arxiv.org/abs/2509.19349>.
- Joel Lehman, Jonathan Gordon, Shawn Jain, Kamal Ndousse, Cathy Yeh, and Kenneth O. Stanley. Evolution through large models, 2022. URL <https://arxiv.org/abs/2206.08896>.
- Pan Li. Dictionaries, not darwin: Set-level selection beats llm evolution in scientific equation discovery, 2026. URL <https://arxiv.org/abs/2607.04108>.
- Yubo Li, Ramayya Krishnan, and Rema Padman. The chain holds, the answer folds: Trace-answer dissociation in reasoning models under adversarial pressure, 2026. URL <https://arxiv.org/abs/2605.29087>.
- Jiaxu Lou and Yifan Sun. Anchoring bias in large language models: An experimental study, 2024. URL <https://arxiv.org/abs/2412.06593>.
- Simon Malberg, Roman Poletukhin, Carolin M. Schuster, and Georg Groh. A comprehensive evaluation of cognitive biases in llms, 2025. URL <https://arxiv.org/abs/2410.15413>.
- Elliot Meyerson, Mark J. Nelson, Herbie Bradley, Adam Gaier, Arash Moradi, Amy K. Hoover, and Joel Lehman. Language model crossover: Variation through few-shot prompting, 2024. URL <https://arxiv.org/abs/2302.12170>.
- Alexander Novikov, Ngân Vũ, Marvin Eisenberger, et al. Alphaevolve: A coding agent for scientific and algorithmic discovery, 2025. URL <https://arxiv.org/abs/2506.13131>.
- OpenAI. OpenAI o3 and o4-mini system card. <https://openai.com/index/o3-o4-mini-system-card/>, 2025.
- Bernardino Romera-Paredes, Mohammadamin Barekatin, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024. doi: 10.1038/s41586-023-06924-6.
- Daniel Scalena, Sara Candussio, Luca Bortolussi, Elisabetta Fersini, Malvina Nissim, and Gabriele Sarti. Beyond the commitment boundary: Probing epiphenomenal chain-of-thought in large reasoning models, 2026. URL <https://arxiv.org/abs/2606.13603>.
- Asankhaya Sharma. Openevolve: an open-source implementation of AlphaEvolve. GitHub repository, <https://github.com/codelion/openevolve>, 2025.
- Parshin Shojaei, Iman Mirzadeh, Keivan Alizadeh, Maxwell Horton, Samy Bengio, and Mehrdad Farajtabar. The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity, 2025. URL <https://arxiv.org/abs/2506.06941>.
- Kevin Sim, Quentin Renau, and Emma Hart. Beyond the hype: Benchmarking llm-evolved heuristics for bin packing, 2025. URL <https://arxiv.org/abs/2501.11411>.
- Charlie Snell, Eric Wallace, Dan Klein, and Sergey Levine. Predicting emergent capabilities by finetuning, 2024. URL <https://arxiv.org/abs/2411.16035>.
- Chang Su, Zhongkai Hao, Zhizhou Zhang, Zeyu Xia, Youjia Wu, Hang Su, and Jun Zhu. Helix: Evolutionary reinforcement learning for open-ended scientific problem solving, 2026. URL <https://arxiv.org/abs/2603.07642>.



- Zhi Rui Tam, Cheng-Kuang Wu, Yi-Lin Tsai, Chieh-Yen Lin, Hung yi Lee, and Yun-Nung Chen. Let me speak freely? a study on the impact of format restrictions on performance of large language models, 2024. URL <https://arxiv.org/abs/2408.02442>.
- Maria Thomas, Kristina Gligoric, and Nihar B. Shah. Mitigating llm-based p-hacking by preregistering for the next llm, 2026. URL <https://arxiv.org/abs/2606.27687>.
- Michelle Vaccaro. Preregistration for experiments with ai agents, 2026. URL <https://arxiv.org/abs/2606.11217>.
- Niki van Stein and Thomas Bäck. Llamea: A large language model evolutionary algorithm for automatically generating metaheuristics, 2025. URL <https://arxiv.org/abs/2405.20132>.
- Haorui Wang, Parshin Shojaee, Kazem Meidani, Kunyang Sun, José Miguel Hernández-Lobato, Teresa Head-Gordon, Jiajun He, Chandan K. Reddy, Chao Zhang, and Yuanqi Du. Towards diverse scientific hypothesis search with large language models, 2026. URL <https://arxiv.org/abs/2606.10587>.
- Yiping Wang, Shao-Rong Su, Zhiyuan Zeng, Eva Xu, Liliang Ren, Xinyu Yang, Zeyi Huang, Xuehai He, Luyao Ma, Baolin Peng, Hao Cheng, Pengcheng He, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, and Yelong Shen. Thetaevolve: Test-time learning on open problems, 2025. URL <https://arxiv.org/abs/2511.23473>.
- Marcus Williams, Hannah Sheahan, Cameron Raymond, Tomek Korbak, Deng Pan, Peilin Yang, Leon Maksin, Ningyi Xie, Phillip Guo, Ian Kivlichan, and Micah Carroll. Predicting llm safety before release by simulating deployment, 2026. URL <https://arxiv.org/abs/2607.07184>.
- Yang Wu, Yifan Zhang, Yiwei Wang, Yujun Cai, Yurong Wu, Yuran Wang, Ning Xu, and Jian Cheng. Answer-centric or reasoning-driven? uncovering the latent memory anchor in llms, 2025. URL <https://arxiv.org/abs/2506.17630>.
- Xia Yang, Xuanyi Zhang, Hao Hu, and Feng Ji. Beyond accuracy: Evaluating strategy diversity in llm mathematical reasoning, 2026. URL <https://arxiv.org/abs/2605.09292>.
- Longfei Yun, Chenyang An, Zilong Wang, Letian Peng, and Jingbo Shang. The price of format: Diversity collapse in llms, 2025. URL <https://arxiv.org/abs/2505.18949>.
- Dong Zhang. Testing frontier large language models’ physics literacy in parallel physical worlds, 2026. URL <https://arxiv.org/abs/2607.00276>.
- Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyang Shi. Verbalized sampling: How to mitigate mode collapse and unlock llm diversity, 2026a. URL <https://arxiv.org/abs/2510.01171>.
- Rui Zhang and Zhichao Lu. Rethinking code similarity for automated algorithm design with llms, 2026. URL <https://arxiv.org/abs/2603.02787>.
- Rui Zhang, Fei Liu, Xi Lin, Zhenkun Wang, Zhichao Lu, and Qingfu Zhang. Understanding the importance of evolutionary search in automated heuristic design with large language models, 2024. URL <https://arxiv.org/abs/2407.10873>.
- Xinhao Zhang, Xi Chen, François Portet, and Maxime Peyrard. What makes an llm a good optimizer? a trajectory analysis of llm-guided evolutionary search, 2026b. URL <https://arxiv.org/abs/2604.19440>.
- Lexin Zhou, Wout Schellaert, Fernando Martínez-Plumed, Yael Moros-Daval, Cèsar Ferri, and José Hernández-Orallo. Larger and more instructable language models become less reliable. *Nature*, 634(8032):61–68, 2024. doi: 10.1038/s41586-024-07930-y.

## A Prompts, verbatim

**A.1 Bare template** ( $\{n\}$  parameterized; the  $N = 13$  instance hashes to 32db485bea625ff9f39f4723ebf1a01f337559a9e2cf567fb486928f71f7f8df):

Pack exactly  $\{n\}$  non-overlapping circles inside the unit square  $[0,1] \times [0,1]$  so that the sum of their radii is as large as possible.  
 MUST hold: exactly  $\{n\}$  circles; every circle fully inside the unit square  $[0,1] \times [0,1]$  ( $x-r \geq 0$ ,  $x+r \leq 1$ , same for  $y$ ); no two circles overlap (distance between centers  $\geq$  sum of the two radii; touching is allowed).  
 Do not write or execute code - construct the packing by reasoning alone.  
 Output ONLY the raw Python list of  $\{n\}$   $[x, y, r]$  lists. No explanation, no code fences, no other text.

**A.2 trace\_v2 template** — identical to A.1 through line 3, then:

First write one line beginning "METHOD:" naming the construction you used.  
 Do not use square brackets anywhere in that line.  
 After the METHOD line, output ONLY the raw Python list of  $\{n\}$   $[x, y, r]$  lists. No explanation, no code fences, no other text after the list.

**A.3 Dispatch wrapper** (both arms; *not* part of the hashed task prompt, hence an instance of the fragment-hashing limitation in §3.1): Do not use any tools. Your entire final message must be the answer and nothing else.

**A.4 Registered prompt hashes.** bare  $N=13$  32db485b...,  $N=17$  8437df75...,  $N=21$  a415425b...,  $N=31$  a664d003...,  $N=35$  3b08c56e...,  $N=37$  2c0a8819...; trace\_v2  $N=13$  a920f1c9...,  $N=21$  91205727...,  $N=31$  bd490b7b.... The  $N=43$  bare hash 1208e7d2... is ledger-only, and the pilot prompt was never hashed (§9).

**A.5 Parser.** Completions are parsed with `ast.literal_eval` after stripping code fences; a completion failing to yield a list of  $N$  numeric triples is recorded `literal_eval_failed` and scored invalid, with the failure mode logged (§6.2 reports parse and geometric failures separately). Radii must be strictly positive; containment and pairwise non-overlap are checked at the two registered tolerances.

## B Deviations and disclosure table

Table 4: Deviations from preregistration.

| Prediction       | Registered rule                                    | As registered? | Outcome / reason                                                                                                                         |
|------------------|----------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| P1–P3            | exact-value point predictions                      | yes            | §3.3, Table 1                                                                                                                            |
| P4 ( $N=37$ )    | converge, 3.0345178                                | yes            | confirmed                                                                                                                                |
| P5 ( $N=35$ )    | truncate, 2.9166667, structural separation         | yes            | 3/4 valid; $n = 4$ , “consistent with”                                                                                                   |
| P-S1...P-S4      | directional tier comparisons                       | yes            | §5                                                                                                                                       |
| P-O1, P-O2, P-O4 | tier comparisons on on-prediction / rival rates    | <b>no</b>      | declared not evaluable post hoc after a 13% validity collapse; the registered disconfirmation (regression toward the trap) did not occur |
| P-O3             | multi-radius fraction $\geq 0.9$ of valid          | yes            | trivially satisfied on 4/4 valid                                                                                                         |
| P-T1             | direction at each $N$ <b>and</b> pooled $p < 0.05$ | yes            | not confirmed, $p = 0.30$                                                                                                                |
| P-T2             | directional, no alpha                              | yes            | confirmed as registered (1/53 vs 2/50); no inferential weight claimed                                                                    |
| P-T3             | directional, no alpha                              | yes            | met as registered; $p = 0.0325$ uncorrected fails Holm; carried by $N = 13$ (§6.4)                                                       |

*continued on next page*

| Prediction                              | Registered rule                                                     | As registered?                  | Outcome / reason                                                                                                                        |
|-----------------------------------------|---------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| P-T4                                    | $\geq 90\%$ of scoreable claims match                               | yes, plus corrected scorer      | 38/41 (93%) original; 54/56 (96.4%) blind, frozen rubric                                                                                |
| P-M1–P-M3                               | modal output = $V(k^*, m)$ at $N = 20/30/41$                        | yes                             | <b>disconfirmed at all three cells</b> — modal outputs are $(k^*+1)$ -grid values (§3.4)                                                |
| P-M4                                    | $N = 57$ modal = $T(8, 57)$ , rival count 0                         | yes                             | confirmed (6/10 on-prediction, 0/10 rival)                                                                                              |
| F-M1 (arm M falsifier, filler branch)   | modal $\neq V(k^*, m)$ at 2+ of 3 converge cells, ties count as NOT | yes, tie-stated in registration | <b>triggered at 3 of 3</b> ; $V(k, m)$ restricted to $m \leq 1$ (§3.4)                                                                  |
| F-M2 (arm M falsifier, truncate branch) | $N = 57$ modal $\neq T(8, 57)$                                      | yes                             | not triggered                                                                                                                           |
| Arm M rejection rule                    | runtime deaths excluded, counted                                    | yes (registered for this arm)   | 3 $N = 41$ invocations excluded, cell reported as 12 of 15                                                                              |
| Falsifier                               | trace_v2 validity $\leq$ bare at 2+ of 3 $N$                        | <b>both readings reported</b>   | <b>triggered</b> under the registered tie-inclusive reading (2/3 cells); not triggered under the strict $<$ the code implemented (§6.3) |
| Ladder inclusion                        | —                                                                   | <b>post hoc</b>                 | <b>opus_alias</b> initially excluded, later included after results known; ladder reported both ways (§5)                                |
| Cap exclusion                           | —                                                                   | <b>not registered</b>           | five records excluded; both rates reported (§3.3)                                                                                       |

## C Corrections ledger

Table 5: Corrections to revision 1. Every claim that changed between the reviewed draft and this one, with the finding that forced it.  
stats = p4\_review\_stats.md, R2 = p4\_review\_reviewer2.md, GECCO = p4\_review\_gecco.md, council = p6\_cruxes.md.

| #  | §           | Revision-1 claim                                                   | Corrected                                                                                                                                                                                                            | Source                       |
|----|-------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1  | Abs, 1      | “recall” a template; “template memorizer”                          | behavioral vocabulary only: emits / concentrates / modal output                                                                                                                                                      | R2 #2; R-E                   |
| 2  | 2.3         | round( $\sqrt{N}$ ) an empirical discovery; “zero free parameters” | a definition — two formalizations of “nearest” coincide on the integers; weakly identified selection among four                                                                                                      | GECCO M8; R2 #24             |
| 3  | 1.1, F1     | penalty “hits exactly zero at the top of each zone”                | zero at $N = 14, 15, 35, 48$ ; <b>0.59% residual at</b> $N = 24$ ; 4 of 22 trap $N$ cost nothing, so branch $\neq$ penalty                                                                                           | GECCO M14; R2 #19            |
| 4  | 1.1, 7      | “published bounds sit above all of them”                           | withdrawn: the $N = 26$ bound (2.63598) <i>is</i> ShinkaEvolve’s, so the LLM systems are the record; table stops at $N = 30$ , leaving the LP abort gate unchecked above it                                          | GECCO M3; R2 #20–21          |
| 5  | 3.3         | cap exclusions “understated validity by 17%”                       | 77.8% vs 70.0% — 7.8 points, 10% relative; exclusion unregistered, both rates reported                                                                                                                               | R2 #13; GECCO m15            |
| 6  | 3.3, 6      | “two parse failures”; “two fraction-literal samples”               | three in each case                                                                                                                                                                                                   | stats F14, F15               |
| 7  | Abs, 1, 3.2 | “predicts the exact sum-of-radix the model will emit”              | predicts the <b>empirical modal output</b> at 7/7 $N$ ; per-sample rate = modal frequency, 56–86% by cell, 46% pooled; round-number baseline 2/69                                                                    | R2 #15; R-E                  |
| 8  | 4.1         | one $a = 3$ sample left the family; rectangle “confirmed”          | two (3.45, 3.5); all 11 valid samples characterized; partial support, CI [21%, 72%], null stated                                                                                                                     | stats F16; R2 #17; GECCO M9c |
| 9  | Abs, 1, 5   | 32/45 = 71%; inversion 71 $\rightarrow$ 100 $\rightarrow$ 13       | 35/45 = 77.8% at $10^{-6}$ , 64.4% at $10^{-9}$ ; inversion 78 $\rightarrow$ 100 $\rightarrow$ 13 (64 $\rightarrow$ 90 $\rightarrow$ 13 strict); matched-cell 83 $\rightarrow$ 100 $\rightarrow$ 13 given separately | stats F1; R2 #3; R-F         |
| 10 | 5           | Sonnet rival 6/30, 3/10 at $N = 31$                                | 5/30 and 2/10 — the 2.75 escape sits $1.47 \times 10^{-3}$ from the rival, inside the window; categorical claims now structural or at $10^{-6}$                                                                      | stats F12; R2 #14            |
| 11 | 1           | “a monotone inversion”                                             | monotone in ambition only; validity rises then collapses                                                                                                                                                             | GECCO m21                    |

continued on next page

| #  | §           | Revision-1 claim                                                                                        | Corrected                                                                                                                                                                                                 | Source                           |
|----|-------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 12 | 6.2         | P-T2 “not confirmed” ( $p = 0.48$ ); P-T3 “confirmed” ( $p = 0.0325$ )                                  | one rule applied symmetrically: P-T2 <b>confirmed as registered</b> (1/53 vs 2/50, no inferential weight); P-T3 met as registered but fragile                                                             | stats F7; R-C                    |
| 13 | Abs, 1, 6.2 | “leaving validity unchanged”                                                                            | no <i>detectable</i> change ( $p = 0.30$ , $n = 60$ /arm, underpowered); the P-T1 direction is entirely parse compliance — 3 vs 0 parse failures, 7 vs 7 geometric                                        | R2 #7–8                          |
| 14 | 6.3         | “the registered falsifier was not triggered”                                                            | <b>triggered at 2 of 3</b> $N$ under the registered $\leq$ ; not under the code’s strict $<$ ; both reported, registered reading authoritative; pilot validity effect attributed to the bundled rewording | stats F8; R-B                    |
| 15 | 6.4         | pilot produced two effects, both died at scale                                                          | a third ran in the P-T3 direction at the same cell (9/10 vs 4/7, $p = 0.16$ ): P-T3 replicated a pilot signal                                                                                             | stats F18                        |
| 16 | 6.5         | 12 claims excluded as “no numeric content”                                                              | regex coverage gaps; 9 of 12 carry checkable dimensions, quoted verbatim in §6.5; worst case $38/53 = 72\%$                                                                                               | stats F9                         |
| 17 | 6.5         | the three mismatches “are all the same case”                                                            | one of three; all three score MATCH under blind adjudication; the two remaining mismatches are alternation mis-descriptions (rows 11, 33)                                                                 | stats F10; R-D                   |
| 18 | 1, 7        | SeaEvo (2604.24372) reports a $\sim 2.636$ packing value                                                | removed: SeaEvo does not evaluate circle packing                                                                                                                                                          | GECCO M4                         |
| 19 | 7           | 2605.29268 reports asymmetric proposal mass in program space                                            | deleted — that paper is on bandit compute allocation; removing it leaves no cited evidence that the anchoring survives the code channel                                                                   | GECCO M5                         |
| 20 | 7           | 2606.13603, 2605.29087 estimate faithfulness by perturbation / probing                                  | both are causal-decoupling results; only 2503.08679 is an estimator                                                                                                                                       | GECCO M6                         |
| 21 | 7, 8        | 2407.10873 and 2604.19440 filed as loop skepticism                                                      | 2407.10873 grounds the <i>importance</i> of evolutionary search — evidence against our substitution, engaged in §8; 2604.19440 recites as “local refiner”                                                 | GECCO M7, m5                     |
| 22 | 1, 7        | AlphaEvolve “2.635”; FunSearch <i>Nature</i> 2023; LMX TELO 2023; AlphaEvolve/ShinkaEvolve unidentified | 2.63586276; <i>Nature</i> 625(7995):468–475, 2024; arXiv 2023 / TELO 2024; both systems identified                                                                                                        | GECCO M3c, m2, m3                |
| 23 | 1, 7        | ThetaEvolve “in the same band”; HELIX ordering unremarked                                               | ThetaEvolve claims new best-known bounds; HELIX sits below ShinkaEvolve while claiming SOTA                                                                                                               | GECCO m10, m11                   |
| 24 | 7           | “EvoDiverse (2606.10587)”; 2505.15392 as numeric anchoring                                              | cited by title, method name unverified; 2505.15392 is general anchoring                                                                                                                                   | GECCO m4, m9                     |
| 25 | 7           | HindsightBench freezes “under SHA-256”; 2607.07184 “files OSF preregistrations”                         | both unsupported by the sources; restated as frozen preregistrations and registered outcome-blinded predictions                                                                                           | GECCO m7, m8                     |
| 26 | 7           | “Mutation Without Variation”: 87% of chains and 93% of mutations                                        | nested, not parallel: in 87% of chains, over 93% of mutations revisit a prior form                                                                                                                        | GECCO m1                         |
| 27 | Abs, 6.1    | “hundreds of invocations”; corpus arithmetic unreconcilable                                             | 231 invocations, itemized in §6.1                                                                                                                                                                         | R2 #30; GECCO M11c               |
| 28 | 3.3, 6      | 18/23 and 35/50 given without stating they are nested subsets                                           | subsets defined in §3.3 and Table 1; full-ledger 41/57 and 2/57 alongside                                                                                                                                 | stats F13; R2 #4; GECCO M11b     |
| 29 | 5           | Sonnet multi-radius “against a Haiku baseline of 13/35”                                                 | dropped; that baseline reproduces at no denominator                                                                                                                                                       | R2 #23a                          |
| 30 | 5           | “no Haiku sample did [this] in 101 invocations”                                                         | 155 weak-tier rows; two exceeded the family best, both by $\sim 10^{-7}$                                                                                                                                  | stats F17; R2 #23b               |
| 31 | 3.1, 9      | dual-tolerance reporting promised, never delivered; “cryptographic prompt hashes”                       | both tolerances in every validity figure; hashes restated as prompt-fragment hashes with coverage gaps itemized above                                                                                     | stats F21; GECCO M12             |
| 32 | all         | five HTML working comments and a merge-provenance header shipped                                        | removed; the two load-bearing ones ( $k = 8$ clipping, $p$ -value convention) promoted into §1.1, Fig 1 and §6                                                                                            | stats F22; R2 #29; GECCO NIT-1–2 |